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KOTA (RAJASTHAN)
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JEE (MAIN+ADVANCED)

# Physical Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

## Solid State

English Medium

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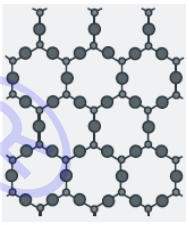
# SOLID STATE

## 1. THE SOLID STATE :

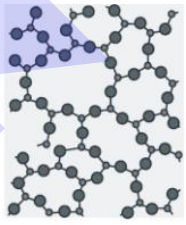
The solid are characterised by incompressibility, rigidity and mechanical strength. The molecules, atoms or ions in solids are closely packed i.e. they are held together by strong forces and can not move about at random. Thus solids have definite volume, shape, slow diffusion, low vapour pressure and possesses the unique property of being rigid.

## 2. AMORPHOUS AND CRYSTALLINE SOLIDS

Solids can be classified as *crystalline* or *amorphous* on the basis of the nature of order present in the arrangement of their constituent particles. A crystalline solid usually consists of a large number of small crystals, each of them having a definite characteristic geometrical shape. In a crystal, the arrangement of constituent particles (atoms, molecules or ions) is ordered. It has ***long range order*** which means that there is a regular pattern of arrangement of particles which repeats itself periodically over the entire crystal. Sodium chloride and quartz are typical examples of crystalline solids. An amorphous solid (Greek *amorphos* = no form) consists of particles of irregular shape. The arrangement of constituent particles (atoms, molecules or ions) in such a solid has only ***short range order***. In such an arrangement, a regular and periodically repeating pattern is observed over short distances only.



(a)



(b)

**Fig. 1.1 :** Two dimensional structure of (a) quartz and (b) quartz glass

Such portions are scattered and in between, the arrangement is disordered. The structures of quartz (crystalline) and quartz glass (amorphous) are shown in Fig. 1.1 (a) and (b) respectively. While the two structures are almost identical, yet in the case of amorphous quartz glass there is no *long range order*. The structure of amorphous solids is similar to that of liquids. Glass, rubber and plastics are typical examples of amorphous solids. Due to the differences in the arrangement of the constituent particles, the two types of solids differ in their properties.

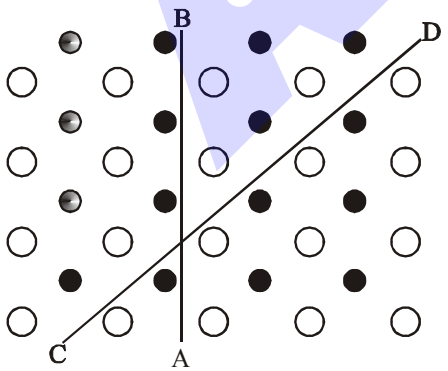



***Fig. 1.2 :*** Anisotropy in crystals is due to different arrangement of particles along different directions.

Crystalline solids have a sharp melting point. On the other hand, amorphous solids soften over a range of temperature and can be moulded and blown into various shapes. On heating they may become crystalline at some temperature. Some glass objects from ancient civilisations are found to become milky in appearance because of some crystallisation. Like liquids, amorphous solids have a tendency to flow, though very slowly. Therefore, sometimes these are called ***pseudo solids*** or ***super cooled liquids.*** Glass panes fixed to windows or doors of old buildings are invariably found to be slightly thicker at the bottom than at the top. This is because the glass flows down very slowly and makes the bottom portion slightly thicker.

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Crystalline solids are ***anisotropic*** in nature, that is, some of their physical properties like electrical resistance or refractive index show different values when measured along different directions in the same crystals. This arises from different arrangement of particles in different directions. This is illustrated in Fig. 1.2. Since the arrangement of particles is different along different directions, the value of same physical property is found to be different along each direction. Amorphous solids on the other hand are ***isotropic*** in nature. It is because there is no *long range* order in them and arrangement is irregular along all the directions. Therefore, value of any physical property would be same along any direction. These differences are summarised in Table below :

**Distinction between Crystalline and Amorphous Solids**

| Property | Crystalline Solids | Amorphous Solids |
|---|---|---|
| Shape | Definite characteristic geometrical shape | Irregular shape |
| Melting point | Melt at a sharp and characteristic temperature | Gradually soften over a range of temperature |
| Cleavage property | When cut with a sharp edged tool, they split into two pieces and the newly generated surfaces are plane and smooth | When cut with a sharp edged tool, they cut into two pieces with irregular surfaces |
| Heat of fusion | They have a definite and characteristic heat of fusion | They do not have definite heat of fusion |
| Anisotropy | Anisotropic in nature | Isotropic in nature |
| Nature | True solids | Pseudo solids or super cooled liquids |
| Order in arrangement of constituent particles | Long range order | Only short range order. |

***Ex.1*** ***Classify the following as amorphous or crystalline solids :***

(a) Polyurethane (b) Napthalene

(c) Benzoic acid (d) Teflon

(e) Potassium nitrate (f) Cellophane

(g) Polyvinyl chloride (h) Fibre glass

(i) Copper

**Sol.** Crystalline : (b) , (c) , (e) , (i)

Amorphous : (a) , (d) , (f) , (g), (h)

**Note :** Polymeric substances are generally amorphous.

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## 3. TYPES OF THE CRYSTALLINE SOLID

| Types of Solid | Constituent Particles | Bonding/ Attractive forces | Examples | Physical Nature | Electrical Conductivity | Melting Point |
|---|---|---|---|---|---|---|
| (1) Molecular Solids (i) Non polar | Molecules | Dispersion or London forces | Ar, $CCl_4$, $H_2$, $I_2$ $CO_2$ | Soft | Insulator | Very low |
| (ii) Polar | | Dipole-dipole | HCl, $SO_2$ | Soft | Insulator | Low |
| (iii) Hydrogen bonded | | Hydrogen bonding | $H_2O$ (ice) | Hard | Insulator | Low |
| (2) Ionic Solids | Ions | Coulombic or electrostatic | NaCl, MgO, ZnS, $CaF_2$ | Hard but brittle | Insulator in solid state but conductors in molten state and in aqueous solutions | High |
| (3) Metallic Solids | Positive ions in a sea of delocalised electrons | Metallic bonding | Fe, Cu, Ag, Mg | Hard but malleable and ductile | Conductors in solid state as well as in molten state | Fairly high |
| (4) Covalent or network Solids | Atoms | Covalent bonding | $SiO_2$ (quartz) SiC, C (diomond) AIN, | Hard | Insulators | Very high |
| | | | C(graphite) | Soft | Conductor | |

***Ex.2*** ***Classify the following solids in different categories based on the nature of intermolecular forces operating in them :***

**(a)** Potassium sulphate ($K_2SO_4$) **(b)** Tin (Sn)

**(c)** Benzene ($C_6H_6$) **(d)** Urea ($NH_2CONH_2$)

**(e)** Ammonia ($NH_3$) **(f)** Water ($H_2O$)

**(g)** Zinc sulphide (ZnS) **(h)** Graphite (C)

**(i)** Rubidium (Rb) **(j)** Argon (Ar)

**(k)** Silicon carbide (SiC) ***(l)*** Bronze

**Sol.** Ionic solids : (a), (g)

Metallic solids : (b), (i), (*l*)

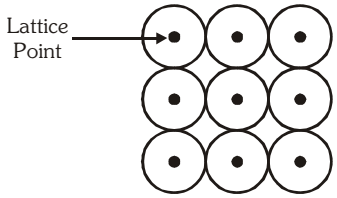
Molecular solids : (c), (d), (e), (f), (j)

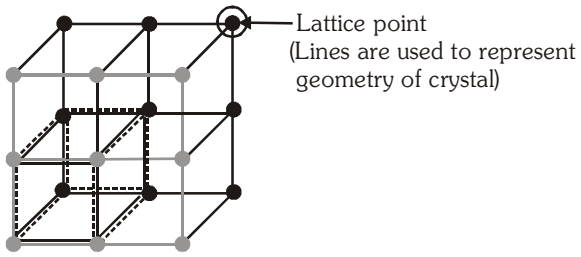
Covalent network solids.: (h), (k)

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## 4.0 SOME BASIC DEFINITION :

## 4.1 SPACE LATTICE (CRYSTAL LATTICE :




The main characteristic of crystalline solids is a regular and repeating pattern of constituent particles. If the three dimensional arrangement of constituent particles in a crystal is represented diagrammatically, in which each particle is depicted as a point, the arrangement is called crystal lattice. Thus, ***a regular three dimensional arrangement of points in space is called a crystal lattice***. A portion of a crystal lattice is shown in Fig. The following are the characteristics of a crystal lattice:

(a) Each point in a lattice is called ***lattice point*** or ***lattice site***.

(b) Each point in a crystal lattice represents one constituent particle which may be an atom, a molecule (group of atoms) or an ion.

(c) Lattice points are joined by straight lines to bring out the geometry of the lattice.

## 4.2. UNIT CELL :

Unit cell is the smallest portion of a crystal lattice which , when repeated in different directions, generates the entire lattice.

A unit cell is characterized by the edge lengths a, b and c along the three edges of the unit cell and the angles $\alpha$, $\beta$ and $\gamma$ between the pair of edges : bc, ca and ab, respectively.

**TYPE OF UNIT CELLS -**

### 4.2.1 Primitive and Centred Unit cells

Unit cells can be broadly divided into two categories, primitive and centred unit cells.

**(a) Primitive Unit Cells (P)**

When constituent particles are present only on the corner positions of a unit cell, it is called as **primitive unit cell.**

**(b) Centred Unit Cells**

When a unit cell contains one or more constituent particles present at positions other than corners in addition to those at corners, it is called a centred unit cell. **Centred unit cells** are of three types:

(i) **Body-Centred Unit Cells (I) :** Such a unit cell contains one constituent particle (atom, molecule or ion) at its body-centre besides the ones that are at its corners.

(ii) **Face-Centred Unit Cells (F) :** Such a unit cell contains one constituent particle present at the centre of each face, besides the ones that are at its corners.

(iii) **End-Centred Unit Cells (E) :** In such a unit cell, one constituent particle is present at the centre of any two opposite faces besides the ones present at its corners.

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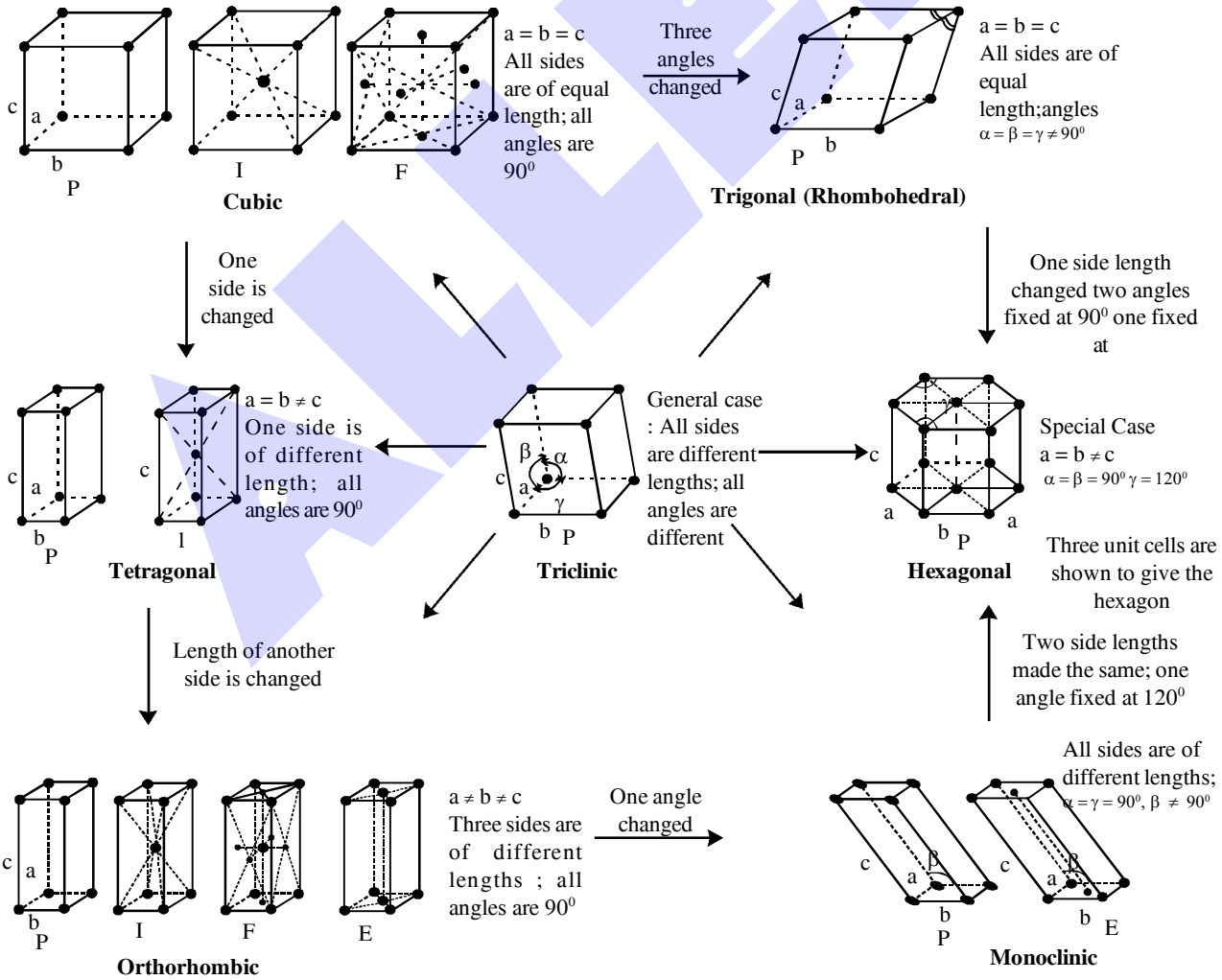
## 4.3. THE SEVEN CRYSTAL SYSTEMS

On the basis of the classification of symmetry, the lattice have been divided into seven systems. These can be grouped into 7 crystal systems. These seven systems with the characteristics of their axes (angles and intercepts) along with some examples of each are given in the following table :

**Seven Primitive Unit cells and their Possible Variations as Centred Unit Cells**

| Crystal system | Possible Variations | Axial distance or edge lengths | Axial angles | Examples |
|---|---|---|---|---|
| Cubic | Primitive, body-centred, Face centre | a = b = c | $\alpha = \beta = \gamma = 90°$ | NaCl, Zinc blende, Cu |
| Tetragonal | Primitive, Body-centred | a = b ≠ c | $\alpha = \beta = \gamma = 90°$ | White tin, $SnO_2$,$TiO_2$, $CaSO_4$ |
| Orthorhombic | Primitive, Body-centred, Face-centred, End-centred | a ≠ b ≠ c | $\alpha = \beta = \gamma = 90°$ | Rhombic sulphur, $KNO_3$, $BaSO_4$ |
| Hexagonal | Primitive | a = b ≠ c | $\alpha = \beta = 90°$ , $\gamma = 120°$ | Graphite, ZnO, CdS, |
| Rhombohedral or Trigonal | Primitive | a = b = c | $\alpha = \beta = \gamma \neq 90°$ | Calcite ($CaCO_3$), HgS (Cinnabar) |
| Monoclinic | Primitive, End-centred | a ≠ b ≠ c | $\alpha = \gamma = 90°$, $\beta \neq 90°$ | Monoclinic sulphur,$Na_2SO_4 \cdot 10H_2O$ |
| Triclinic | Primitive | a ≠ b ≠ c | $\alpha \neq \beta \neq \gamma \neq 90°$ | $K_2Cr_2O_7$,$CuSO_4.5H_2O$,$H_3BO_3$ |

## 4.4. BRAVAIS LATTICES : There are 14 Bravais lattices :




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### 4.5. CO-ORDINATION NUMBER :

The number of nearest particles around a specific particle in a given crystalline substance is called ***co-ordination*** number.

### 4.6. PACKING EFFICIENCY OR PACKING DENSITY (P.E.) :

Packing efficiency is defined as the ratio of volume occupied by the constituent particles to the total volume of the crystalline substance.

$$\text{P.E.} = \frac{\text{Volume occupied by particles present in a crystal}}{\text{Volume of crystal}}$$

$$\text{P.E.} = \frac{\text{Volume occupied by particles present in unit cell}}{\text{Volume of Unit Cell}}$$

$$\text{P.E.} = \frac{Z \times (4/3)\pi r^3}{V}$$ , where Z = number of atoms present in unit cell

### 4.7. *DENSITY OF THE CRYSTAL :*

$$\text{Density of crystal} = \text{Density of an unit cell} = \frac{\text{Mass of unit cell}}{\text{Volume of unit cell}}$$

Mass of the unit cell = Number of particles present in a unit cell $\times$ Mass of one particles = Z $\times$ m

$$\text{But mass of one particles (m)} = \frac{\text{Particle mass}}{\text{Avogadro Number}} = \frac{M}{N_A}$$

$$\text{Mass of an unit cell} = Z \times \frac{M}{N_A}$$

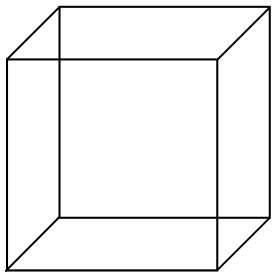
$$\text{Density of an unit cell} = \frac{Z \times \frac{M}{N_A}}{V}$$

$\therefore$ **Density of Crystal, d = Density of an unit cell** $= \frac{\mathbf{Z \times M}}{\mathbf{V \times N_A}}$ **g cm$^{-3}$**

## 5.0 ANALYSIS OF CUBIC CRYSTAL :

### 5.1. GEOMETRY OF A CUBE

Number of corners = 8
Number of faces = 6
Number of edges = 12
Number of cube centre = 1
Number of cube diagonals = 4
Number of face diagonals = 12

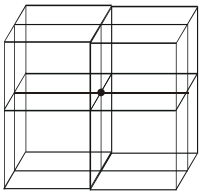


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## 5.2 CONTRIBUTION OF A CONSTITUENT PARTICLE AT DIFFERENT SITES OF CUBE :

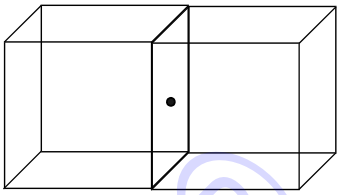
**5.2.1** A corner of a cube is common in 8 cubes.

So $\frac{1}{8}$ th part of a particle is present at this corner of cube.



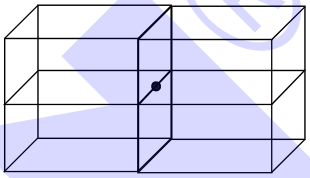
**5.2.2** A face of a cube is common is 2 cubes.

So $\frac{1}{2}$ th part of a particle is present at the face of a cube.



**5.2.3** An edge of a cube is common in four cubes,

so $\frac{1}{4}$ th part of particle is present at the edge of a cube

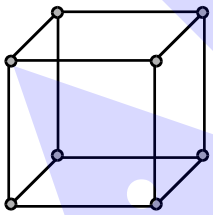


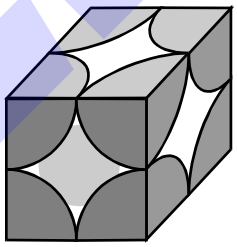
## 5.3 TYPE OF CUBIC UNIT CELL :

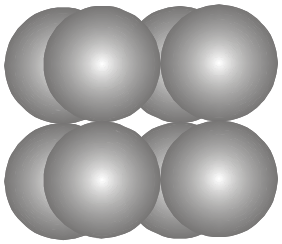
### 5.3.1 Simple/Primitive/Basic Unit cell (Simple cubic, SC) ;

A unit cell having lattice point only at corners called as primitive or simple unit cell. In this case there is one particle at each of the eight corners of the unit cell.

Considering a particles at one corner as the centre, it will be found that this atom is surrounded by six equidistant neighbours (particles) and thus the co-ordination number will be six. If 'a' is the side of the unit cell, then the distance between the nearest neighbours shall be equal to 'a'.



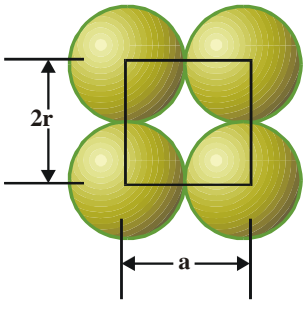




**(a) Relationship between Edge length 'a' and Particle radius 'r' :-**

$a = 2r$

**i.e. ,** $\boxed{r = \frac{a}{2}}$




**(One face of SC)**

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**(b) Number of particles present in unit cell (Z) :** In this case one particle lies at each corner. Hence simple cubic unit cell contains a total of $\frac{1}{8}\times 8 = 1$ particle /unit cell.

**(c) Packing efficiency (P. E.) :**

$$\text{P.E.} = \frac{\text{Volume occupied by particles present in unit cell}}{\text{Volume of unit cell}} = \frac{Z\times\frac{4}{3}\pi r^3}{V} \quad \left[\text{Volume of atom} = \frac{4}{3}\pi r^3\right]$$

$$\text{P.E.} = \frac{1\times\frac{4}{3}\times\pi\times\left(\frac{a}{2}\right)^3}{a^3} = \frac{\pi}{6} = 0.52 \text{ or } 52\% \qquad \left[r = \frac{a}{2} \text{ and } V = a^3, Z = 1\right]$$

In SC, 52% of total volume is occupied by particles.

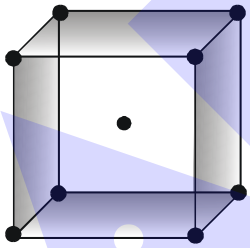
$\therefore$ Void space $\approx$ (100 – 52)= 48 %

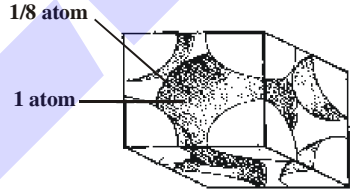
**(d) Coordination number (CN)**

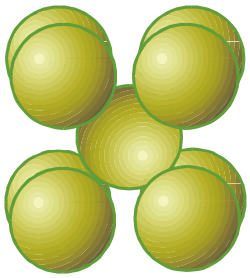
| Nearest neighbour | Distance | Number |
|---|---|---|
| 1 | a | 6 |
| 2 | $\sqrt{2}$ a | 12 |
| 3 | $\sqrt{3}$ a | 8 |

### 5.3.2. Body Centred Cubic unit cell (BCC):

A unit cell having lattice point at the body centre in addition to the lattice points at every corner is called as body centered unit cell. Here the central particle is surrounded by eight equidistant particles and hence the co-ordination number is eight. The nearest distance between two particles will be $\frac{a\sqrt{3}}{2}$





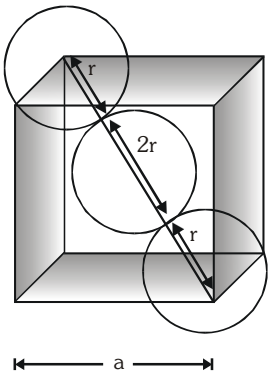



**(a) Relationship between edge length 'a' and particle radius 'r' :**

In BCC, along cube diagonal all particles touches each other and the length of cube diagonal is $\sqrt{3}a$ .

So, $\sqrt{3}a = 4r$

i.e. $r = \frac{\sqrt{3}a}{4}$




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**(b) Number of particle present in unit cell (Z):**

$$Z = \left(\frac{1}{8}\times 8\right) \quad + \quad (1\times 1) = 1 + 1 = 2 \text{ particles/unit cell.}$$

(Corner) (Body centre)

In this case one particle lies at the each corner of the cube. Thus contribution of the 8 corners is $\left(\frac{1}{8}\times 8\right) = 1$, while that of the body centred is 1 in the unit cell. Hence total number of particles per unit cell is 1 + 1 = 2

**(c) Packing efficiency :**

$$\text{P.E.} = \frac{Z\times\frac{4}{3}\pi r^3}{V} = \frac{2\times\frac{4}{3}\times\pi\left(\frac{\sqrt{3}a}{4}\right)^3}{a^3} = \frac{\sqrt{3}\pi}{8} = 0.68 \text{ or } 68\% \qquad [Z = 2, r = \frac{\sqrt{3}a}{4}, V = a^3]$$

In BCC, 68% of total volume is occupied by particles.

$\therefore$ Void space = 100 – 68 = 32 %

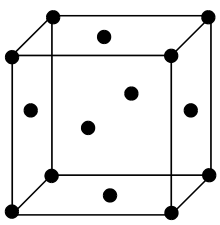
**(d) Coordination number (CN)**

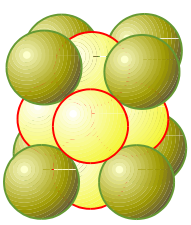
| Nearest neighbour | Distance | Number |
|---|---|---|
| 1 | $\sqrt{3}\frac{a}{2}$ | 8 |
| 2 | a | 6 |
| 3 | $\sqrt{2}a$ | 12 |

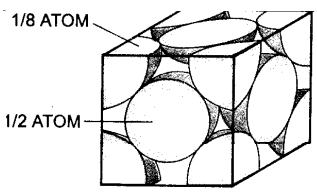
### 5.3.3 Face Centred Cubic unit cell (FCC):

A unit cell having lattice point at every face centre in addition to the lattice point at every corner called as face centred unit cell.

In this case there are eight particles at the eight corners of the unit cell and six particles at the centre of six faces. Considering a particle at the face centre as origin, it will be found that this face is common to two cubes and there are twelve points surrounding it situated at a distance which is equal to half the face diagonal of the unit cell. Thus the co-ordination number will be twelve and the distance between the two nearest particles will be $\frac{a}{\sqrt{2}}$.







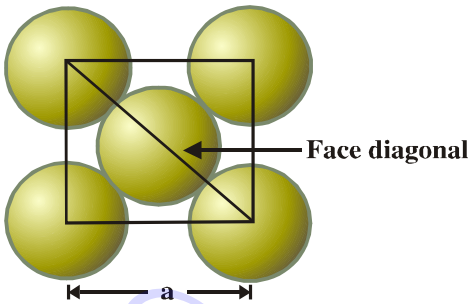

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**(a) Relationship between edge length 'a' and atomic radius 'r' :**

In FCC, along the face diagonal all atoms touches each other and the length of face diagonal is $\sqrt{2}a$.

So $4r = \sqrt{2}a$

i.e. $r = \frac{a}{2\sqrt{2}}$




**(b) Number of particles per unit cell : (Z)**

$Z = \underset{\text{Corner}}{\left(\frac{1}{8}\times 8\right)} + \underset{\text{faces}}{\left(6\times\frac{1}{2}\right)} = 1 + 3 = 4$ particles/unit cell

In this case, one particle lies at the each corner of the cube and one particle lies at the centre of each face of the cube. It may noted that only 1/2 of each face sphere lie within the unit cell and there are six such faces. The total contribution of 8 corners is $\left(\frac{1}{8}\times 8\right) = 1$, while that of 6 face centred particles is $\left(\frac{1}{2}\times 6\right) = 3$ in the unit cell.

Hence, total number of particles per unit cell is $1 + 3 = 4$

**(c) Packing efficiency :**

$$\text{P.E.} = \frac{Z\times\frac{4}{3}\pi r^3}{V} \qquad [Z = 4,\ r = \frac{a}{2\sqrt{2}},\ V = a^3]$$

$$= \frac{4\times\frac{4}{3}\pi\times\left(\frac{a}{2\sqrt{2}}\right)^3}{a^3} = \frac{\pi}{3\sqrt{2}} = 0.74 \text{ or } 74\%$$

In FCC, 74% of total volume is occupied by particles. This is maximum for crystals having identical particles.

$\therefore$ Void space = 100 – 74 = 26 %

**(d) Coordination number (CN)**

| Nearest neighbour | Distance | Number |
|---|---|---|
| 1 | $\frac{a}{\sqrt{2}}$ | 12 |
| 2 | a | 6 |
| 3 | $\sqrt{\frac{3}{2}}a$ | 24 |

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## 5.4 SUMMARY OF CUBIC CRYSTAL :

| Unit cell | No. of particles per unit cell (Z) | 2r = | CN | Volume occupied by particles (%) |
|---|---|---|---|---|
| Simple cube (SC) | 1 | a | 6 | 52 |
| Body centred cube (BCC) | 2 | $\frac{a\sqrt{3}}{2}$ | 8 | 68 |
| Face centred cube (FCC) | 4 | $\frac{a}{\sqrt{2}}$ | 12 | 74 |

***Ex.3. An element (atomic mass = 60) having face centred cubic crystal has a density of 6.23 g cm$^{-3}$. What is the edge length of the unit cell ?***
***(Avogadro constant, $N_A$ = 6.02 × 10$^{23}$ mol$^{-1}$)***

***Sol.*** *Since element has fcc structure hence there are 4 atoms in a unit cell ( Z = 4), Atomic mass is 60 (M = 60), $N_A$ = 6.02 × 10$^{23}$ and d = 6.23 g cm$^{-3}$.*

$$\because \quad d = \frac{Z \times M}{V \times N_A}$$

$$or \quad V = \frac{Z \times M}{d \times N_A} = \frac{(4)\left(60\ g\ mol^{-1)}\right)}{\left(6.23\ g\ cm^{-3)}\right)\left(6.02 \times 10^{23}\ mol^{-1}\right)}$$

$$= 64 \times 10^{-24}\ cm^3$$

*Let $\ell$ be the length of the edge of the unit cell.*

$$\therefore \quad \ell^3 = V = 64 \times 10^{-24}\ cm^3 \quad or \quad \ell = 4.0 \times 10^{-8}\ cm$$

***Ex.4. The density of Al is 5.4 g/cm$^3$. If it crystallise in fcc lattice, determine its atomic radius.***

***Sol.*** $d = \frac{ZM}{N_A.V} \Rightarrow 5.4 = \frac{4 \times 27}{N_A \times (2\sqrt{2}r)^3}$

$\therefore$ r = 1.136 × 10$^{-8}$ cm = 1.136 Å

***Ex.5. A solid crystallises in cubic crystal in which 'X' atoms occupy all the corners and body centres and 'Y' atoms occupy all the face-centres. What is the simplest formula of solid?***

***Sol.*** $Z_X = 8 \times \frac{1}{8} + 1 = 2$

$Z_Y = 6 \times \frac{1}{2} = 3$

$\therefore$ *Simplest formula of solid = $X_2Y_3$*

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## 6. CLOSE PACKING OF IDENTICAL SOLID SPHERES

The solids which have non-directional bonding, their structures are determined on the basis of geometrical consideration. For such solids, it is found that the lowest energy structure is that in which each particle is surrounded by the greatest possible number of neighbours. In order to understand the structure of such solids, let us consider the particles as hard sphere of equal size in three directions. Although there are many ways to arrange the hard spheres but the one in which maximum available space is occupied will be economical which is known as **closed packing**.

To clearly understand the packing of these spheres, the packing can be categorised as :

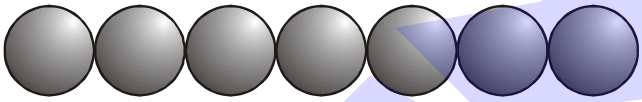
**(i)** Close packing in one dimension.

**(ii)** Close packing in two dimension.

**(iii)** Close packing in three dimension.

### 6.1 CLOSE PACKING IN ONE DIMENSION :

In one dimension , only one arrangement of spheres is possible as shown in fig.



**Close packing of spheres in one dimension**

Each sphere is touched by other two spheres, hence coordination numbers of packing is two.

### 6.2 CLOSE PACKING IN TWO DIMENSION :

Two possible types of two dimensional packing are

(i) Square close packing in two dimension.

(ii) Hexagonal close packing in two dimension.

**(i) Square close packing in two dimension :**

When two rows are placed in such a manner, that spheres of one row are placed immediately below of the other, the resulting packing is called two dimensional square close packing.

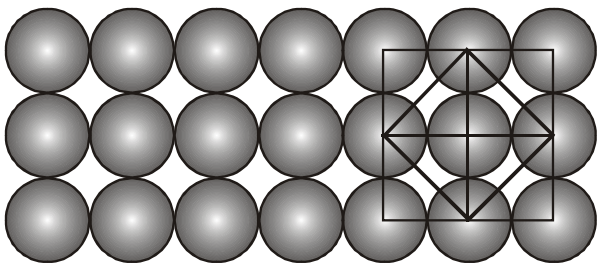
(i) Since all the rows are identical, the packing is called AAA ..... type packing.

(ii) Each sphere is touched by four other, hence coordination number is four.

(iii) If centres of spheres are connected, square cells are formed , hence it also called two dimensional square packing.

(iv) This type of packing is not very effective in terms of utilisation of space.

(v) Packing efficiency $= \frac{1 \times \pi r^2}{a^2} = \frac{1 \times \pi (a/2)^2}{a^2} = \frac{\pi}{4} = 0.78$ or 78%



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**(ii) Hexagonal close packing in two dimension :**

If various one dimensional close pack rows are placed in such a way that spheres of top row fits in depression of bottom row spheres, the resulting packing is called two dimensional hexagonal close packing.

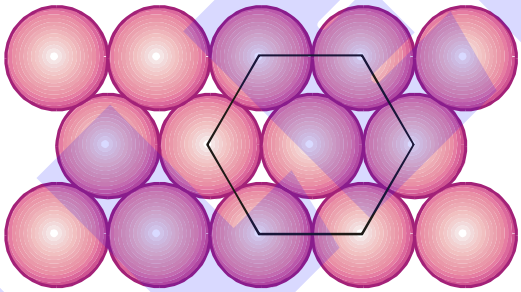
(i) Every third row sphere comes exactly at top of first row sphere, hence the packing is called ABABAB ..... packing.

(ii) If centres are joined, hexagonal unit cells are formed. Hence this is called two dimensional hexagonal close packing.

(iii) This packing is most efficient in utilising space in two dimensional arrangement.

(iv) Each sphere is touched by six other, hence coordination number is six.

(v) Packing efficiency $= \dfrac{3\times\pi\left(\frac{a}{2}\right)^2}{\frac{a^2\sqrt{3}}{4}\times 6} = \dfrac{\pi}{2\sqrt{3}} = 0.90$ or 90%



## 6.3 CLOSE PACKING IN THREE DIMENSIONS :

When two dimensional packing structure are arranged one above the other, depending upon type of two dimensional arrangement in a layer, and the relative positions of spheres in above or below layer, various types of three dimensional packing results. To define 3-D lattice, six lattice parameters are required - 3 edge lengths & 3 angles.

**(i)** Simple cubic packing (A A A A .....)

**(ii)** Hexagonal close packing (AB AB AB .....)

**(iii)** Cubic close packing or face centred cubic (ABC ABC...)

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### 6.3.1 Three dimensional close packing from square two dimensional packing (Simple cubic packing in three dimension )

The two dimensional square close packed layer are placed, in such a manner that spheres in each layer comes immediately on top of below layer, simple cubic packing results. Important points :

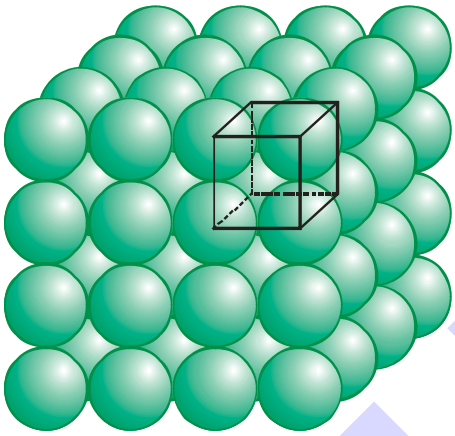
**(i)** Spheres all aligned vertically and horizontally in all directions.

**(ii)** The unit cell for this packing is simple cubic unit cell.

**(iii)** In this packing, only 52% of available space is occupied by spheres.

**(iv)** Each sphere is in contact will six spheres and hence coordination number is 6.

**(v)** Packing efficiency $= \frac{1 \times \frac{4}{3}\pi r^3}{(2r)^3} = \frac{\pi}{6} \approx 0.52$

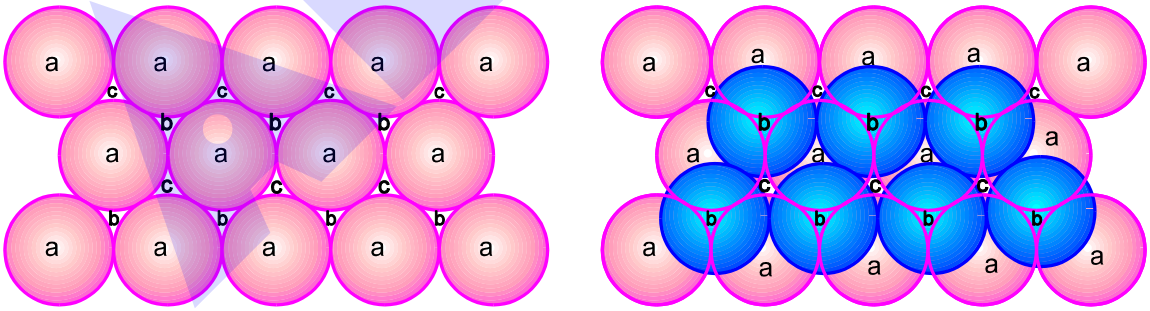


Simple cubic lattice formed by
A A A ... arrangement

Coordination number = 6
First neighbour = 6 at a distance = a
Second neighbour = 12 at ($\sqrt{2}$a) distance
Third neighbour = 8 at ($\sqrt{3}$a) distance

### 6.3.2 Three dimensional close packing from hexagonal two dimensional packing :

In hexagonal close packing, there are two types of the voids (open space or space left) which are divided into two sets 'b' and 'c' for convenience. The spaces marked 'c' are curved triangular spaces with tips pointing upwards whereas spaces marked 'b' are curved triangular spaces with tips pointing downwards.




Now we extend the arrangement of spheres in three dimensions by placing second close packed layer (hexagonal close packing) (B) on the first layer (A). The spheres of second layer may placed either on space denoted by 'b' or 'c'. It may be noted that it is not possible to place spheres on both types of voids (i.e. b and c). Thus half of the voids remain unoccupied by the second layer. The second layer also have voids of the types 'b' and in order to build up the third layer, there are following ways :

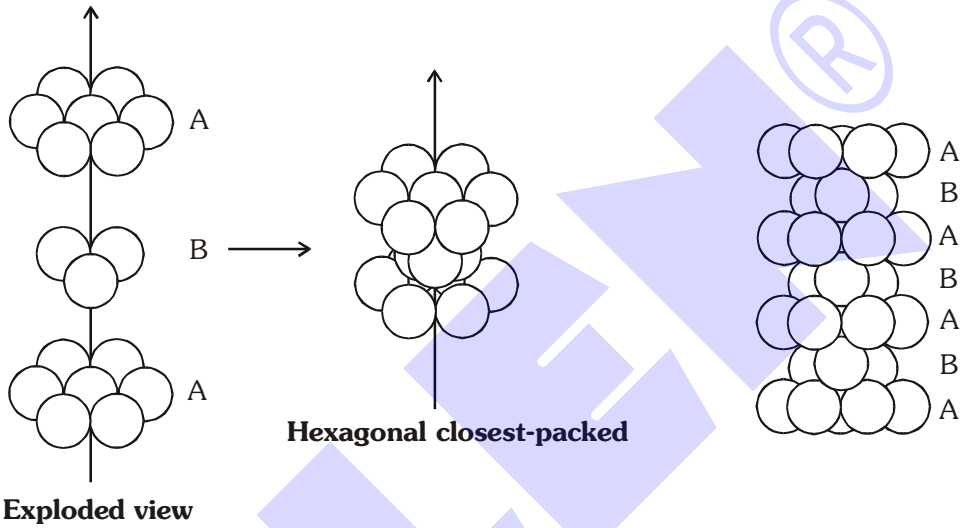
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**(I) Hexagonal close packing (HCP)**

**(i)** In one way, the spheres of the third layer lie on the spaces of second layer (B) in such a way that they lie directly above those in the first layer(A). In other words we can say that the third layer becomes indentical to the first layer. Such arrangement is called AB AB AB ....type packing or hexagonal close packing (hcp).

**(ii)** Maximum possible space is occupied by spheres.

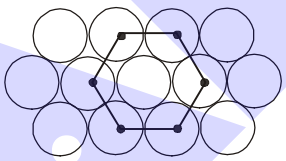
**(iii)** Each sphere is touched by 12 other spheres in 3D (6 is one layer, 3 in top layer and 3 in bottom) and hence the coordination number is 12.

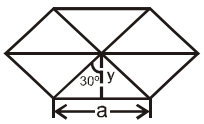



**(iv) Packing efficiency of HCP units**

**Relation between a, b, c and R :**

$a = b = 2R$



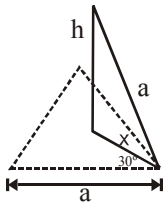



$\tan 30^\circ = \frac{a}{2 \times y}$ So $y = \frac{a \times \sqrt{3}}{2 \times 1} = \frac{\sqrt{3}}{2} a.$

$$\text{Base Area} = 6\left[\frac{1}{2} a \times \frac{\sqrt{3}}{2} a\right] = \frac{6\sqrt{3}a^2}{4}$$

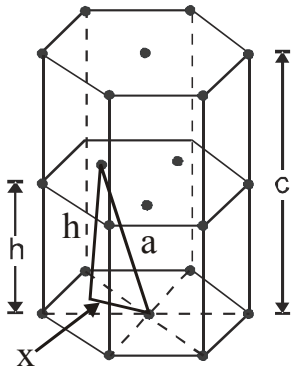
**Calculation of c :**

$\cos 30^\circ = \frac{a}{2 \times x}$ $x = \frac{2a}{2 \times \sqrt{3}} = \frac{a}{\sqrt{3}}$




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Applying pythogoras theorem : $x^2 + h^2 = a^2$




So $h^2 = a^2 - x^2 = a^2 - \frac{a^2}{3} = \frac{2}{3}a^2$

$h = \sqrt{\frac{2}{3}}\,a$ so $c = 2h = 2\sqrt{\frac{2}{3}}\,a$

So volume of hexagon = area of base × height

$$= \frac{6.\sqrt{3}}{4} \times a^2 \times 2\sqrt{\frac{2}{3}}\,a = \frac{6.\sqrt{3}}{4} \times (2R)^2 + 2\sqrt{\frac{2}{3}} \times (2R) = 24\sqrt{2}\,R^3$$

**(v) Effective no. of particles (Z)**

$$Z = 3 + 2 \times \frac{1}{2} + 12 \times \frac{1}{6} = 3 + 1 + 2 = 6.$$

It must be noted that all three spheres of 'B' layer are not exactly inside the unit cell. But the contribution of three spheres are taken because the same volume of other spheres in that layer is also inside the unit cell.

**(vi) Packing efficiency** $= \frac{6 \times \frac{4}{3}\pi R^3}{24\sqrt{2}\,R^3} = \frac{\pi}{3\sqrt{2}} = 0.74$ or 74%.

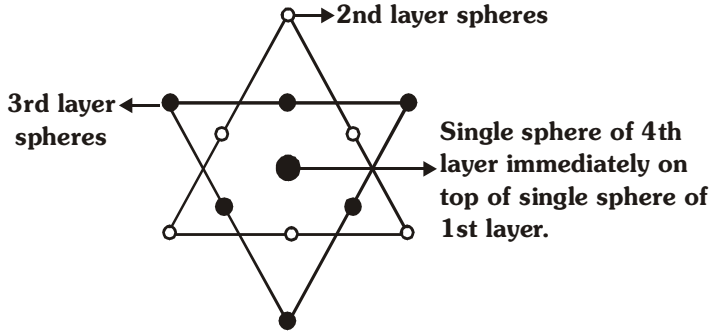
**(vii) Density (d)** $= \frac{\text{mass}}{\text{volume}} = \left[\frac{Z \times M}{N_A \times \text{volume}}\right]$

**(II) Cubic close packing (CCP) or face centered cubic (FCC)**

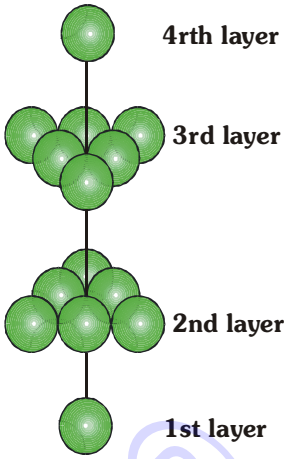
In second way, the spheres of the third layer (C) lie on the second layer (B) in such a way that they lie over the unoccupied spaces 'C' of the first layer(A). If this arrangement is continued in the same order every fourth layer becomes identical to the first. Such arrangement of particle is called ABC ABC ABC.... or cubic close packing (ccp) or face centered cubic (fcc).

It may be noted that in ccp (or fcc) structures, each sphere is surrounded by 12 spheres hence the coordination number of each sphere is 12. The spheres occupy 74% of the total volume and 26% of is the empty space in both (hcp and ccp) structures.

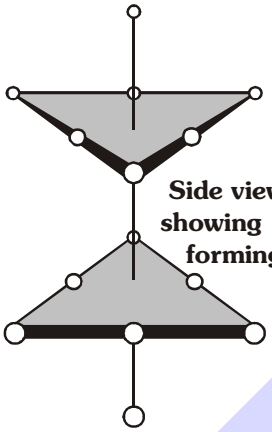
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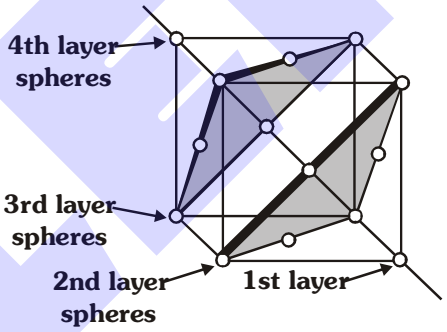

Open structure show top view of 14 spheres forming ccp packing.




Side view exploded space filling diagram showing 14 selected atom forming ccp packing



Side view open structure showing 14 selected atoms forming ccp packing




fcc unit cell showing 14 spheres

**(i) Relation between 'a' and 'R' :**

$a \neq 2R$

$\sqrt{2}a = 4R$ (sphere are touching along the face diagonal)

**(ii) Effective no. of particles per unit cell (Z)**

$$Z = \frac{1}{8}\times 8 + \frac{1}{2}\times 6 = 4$$

**(iii) Packing fraction :**

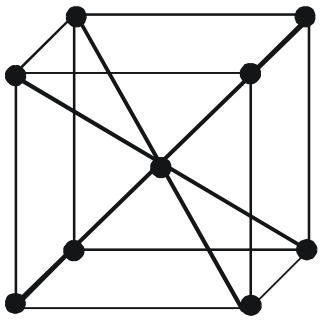
$$\text{P.F.} = \frac{4\times\frac{4}{3}\pi R^3}{4\times 4\times 4\ R^3}\times\sqrt{2}\times 2 = \frac{\pi}{3\sqrt{2}} = 0.74 \text{ or } 74\%$$

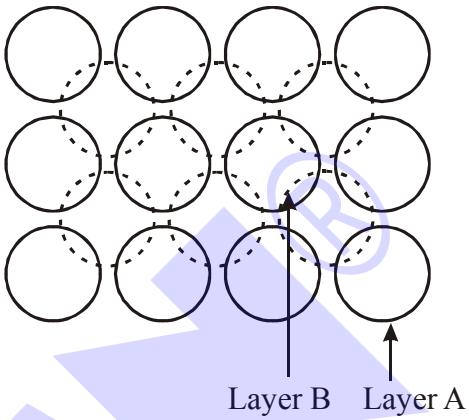
**(iv) Density (d)** $= \frac{Z\times M}{N_A \times a^3}$

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### 6.3.3 Body centred cubic (bcc):

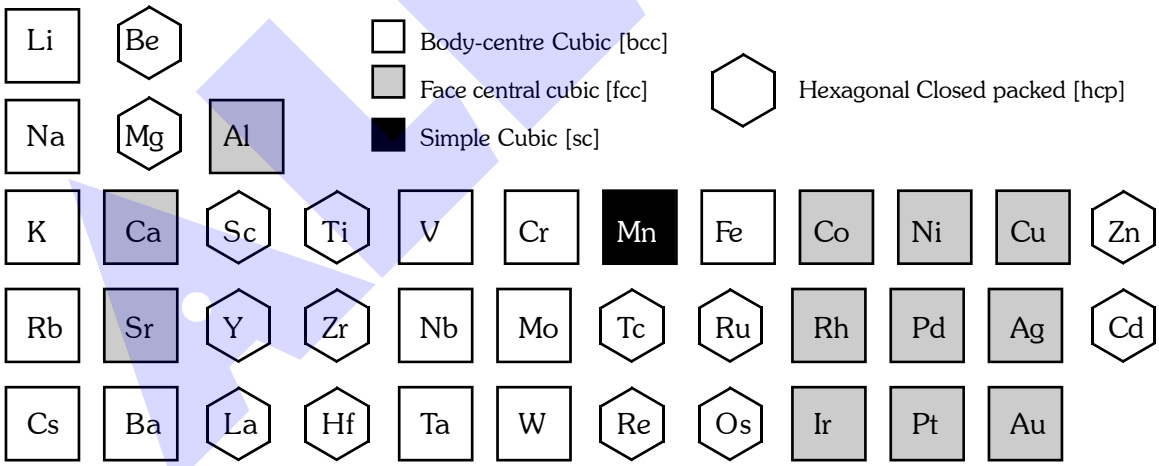
There is another possible arrangement of packing of spheres known as body centred cubic (bcc) arrangement. This arrangement is observed in square close packing in which there is suitable space between the spheres in each layer. In bcc arrangement, the spheres of the second layer lie at the space (hollows or voids) in the first layer.






Thus each sphere of the second layer touches four spheres of the first layer. Now spheres of the third layer are placed exactly about the spheres of first layer. In this way each sphere of the second layer touches eight spheres (four of 1st layer and four of IIIrd layer). Therefore coordination number of each sphere is 8 in bcc sturcture. The spheres occupy 68% of the total volume 32% of the volume is the empty space.

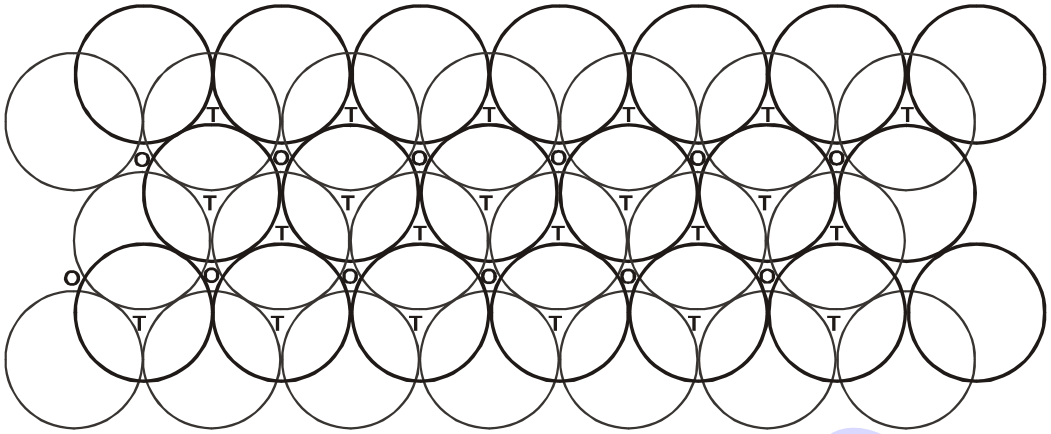
## 7. STRUCTURES OF VARIOUS ELEMENTS




## 8. INTERSTICES OR VOIDS OR HOLES IN CRYSTALS

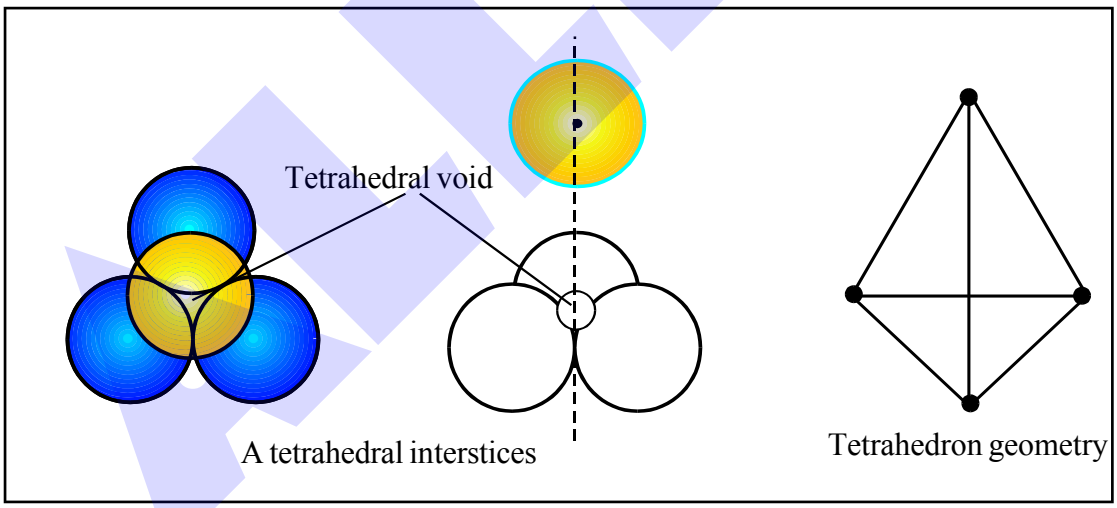
It has been shown that the particles are closely packed in the crystals even than there is some empty space left in between the spheres. This is known as interstices (or interstitial site or hole or empty space or voids). In three dimentional close packing (CCP & HCP) the interstices are of two types : (i) tetrahedral voids and (ii) octahedral voids.

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**A stack of two layers of close packed spheres and voids generated in them. T = Tetrahedral void; O = Octahedral void**

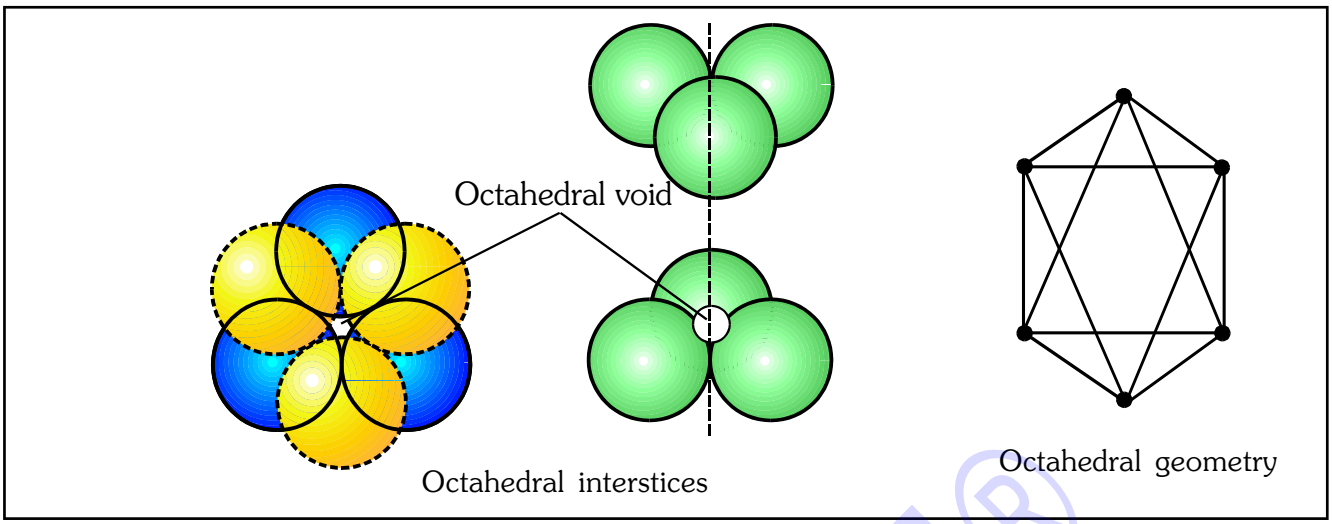
**(i) Tetrahedral voids :** We have seen that in hexagonal close packing (hcp) and cubic close packing (ccp), each sphere of second layer touches with three spheres of first layer. Thus, they leave a small space in between which is known as **tetrahedral site or interstices**. In another words, the vacant space between 4 touching spheres is called as tetrahedral void. Since a sphere touches three spheres in the below layer and three spheres in the above layer hence there are two tetrahedral sites associated with one sphere. It may by noted that a tetrahedral site does not mean that the site is tetrahedral in geometry but it means that this site is surrounded by four spheres and the centres of these four spheres lie at the apices of a regular tetrahedron.




**(ii) Octahedral voids :** Hexagonal close packing (hcp) and cubic close packing (ccp) also form another type of interstices (or site), which is called octahedral site (or interstices). In another words, the vacant space between 6 touching spheres is called as octahedral void.

In the figure two layers of close packed spheres are shown. The spheres of first layer are shown by full circles while that of second layer by dotted circles. Two triangles are drawn by joining the centres of three touching spheres of both the layers.

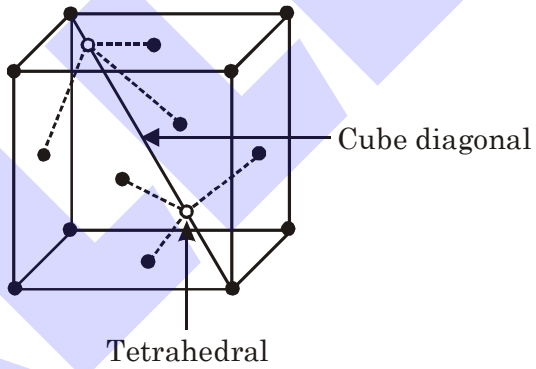
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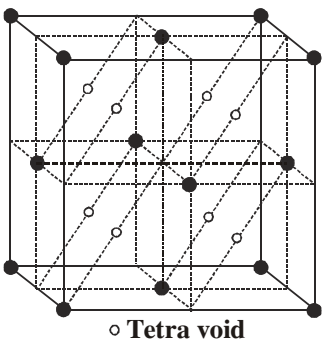

The apices of these triangles point are in opposite directions. On super imposing these triangles on one another, an octahedral site is created. It may be noted that an octahedral site does not mean that the hole is octahedral in shape but it means that this site is surrounded by six nearest neighbour lattice points arranged octahedrally.

## 8.1 POSITIONS OF TETRAHEDRAL VOIDS IN AN FCC UNIT CELL :

In FCC, one corner and its three face centred atom of faces meeting at that corner form a tetrahedral void. In FCC, two tetrahedral voids are obtained along one cube diagonal. So in FCC 8 tetrahedral voids are present.




Alternatively, the centre of tetrahedral void is located on the centre of body diagonal of each small cube of volume $\left(\frac{a^3}{8}\right)$.




Total number of particles per unit cell $= \frac{1}{2} \times 6 + 8 \times \frac{1}{8} = 4$

Total number of tetrahedral void = 8

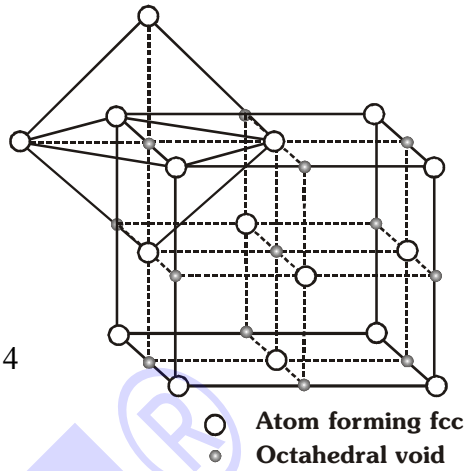
$\therefore$ Effective number of tetrahedral void per particle = 2.

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## 8.2 POSITION OF OCTAHEDRAL VOID IN FCC UNIT CELL :

Position of octahedral void is at mid-point of each edge (total 12 edges in a cube) and at the centre of cube.

Each octahedral void located at mid point of edge contributes 1/4 to the unit cell. The octahedral void situated at the centre contributes 1.




In FCC, total number of octahedral voids are

$$\underset{\text{(Cube centre)}}{(1 \times 1)} + \underset{\text{(edge)}}{(12 \times \frac{1}{4})} = 1 + 3 = 4$$

In FCC, number of particles = 4

$\therefore$ Effective number of octahedral voids per particle = 1

## 8.3 POSITION OF TETRAHEDRAL AND OCTAHEDRAL VOID IN HCP:

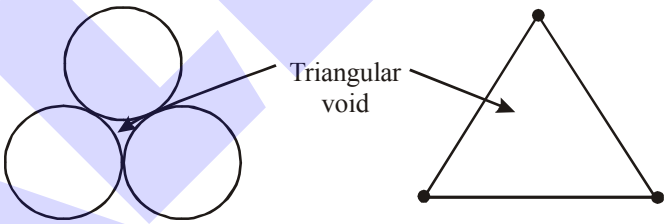
(i) Above & below each sphere, there is a tetrahedral void, hence number of tetrahedral void per unit cell = 12

(ii) Other than TV, all voids are OV & number of OV per unit cell = 2 × 3 = 6

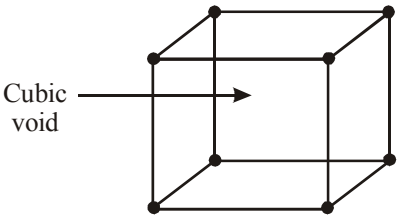
***Note : In any closed packing of maximum efficiency, there are two TV & one OV per particle***

## 8.4 Triangular and cubic voids :

**(i) Triangular void :** It is formed by three spheres in a plane.




**(ii) Cubic void :** It is formed in simple cubic unit cell by eight spheres at the corners of cube.

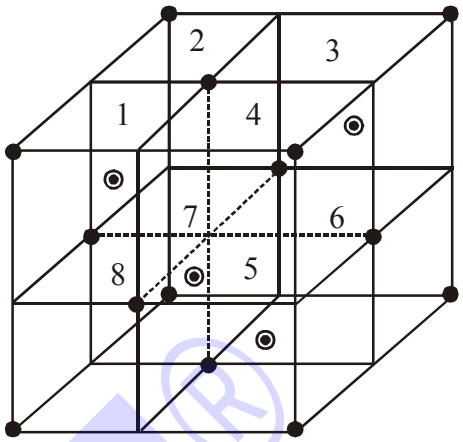



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## 9. CRYSTAL OF DIAMOND

The crystal is FCC for C atom & alternate tetrahedral voids are also occupied by carbon atoms.

* $Z = \frac{1}{8}\times 8 + 6\times\frac{1}{2} + 4 = 8$
* CN = 4
* $\frac{\sqrt{3}a}{4} = 2r = d_{C-C}$
* Number of C–C bonds/unit cell = 4 × 4 =16
* Number of C–C bonds/C-atom = $\frac{16}{8} = 2$
* $PE = \frac{8\times\frac{4}{3}\pi r^2}{\left(\frac{8r}{\sqrt{3}}\right)^3} = \frac{\sqrt{3}\pi}{16} \simeq 0.34 \text{ or } 34\%$




***Ex.6** A solid crystallises in close packing for 'P' atoms and 25% of tetrahedral voids are occupied by 'Q' atoms. What is the simplest formula of solid ?*

***Sol.*** $Z_P = 1(\text{say})$

$Z_Q = \frac{25}{100}\times 2 = \frac{1}{2}$

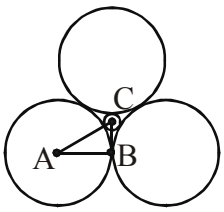
$\therefore$ Simplest formula of solid = $PQ_{1/2} = P_2Q$

***Ex.7** Calculate the radius (r) of largest sphere which may be fitted in the (i) triangular voids (ii) tetrahedral voids (iii) octahedral voids (iv) cubic voids made by identical spheres of radius, 'R', without disturbing the crystal.*

***Sol.*** (i) AB = R

AC = R + r

$\angle BAC = 30°$

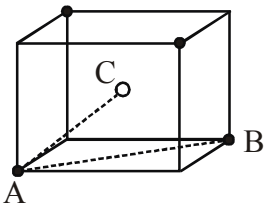



Now, $\cos 30° = \frac{AB}{AC} \Rightarrow \frac{\sqrt{3}}{2} = \frac{R}{R+r}$

$r = \left(\frac{2}{\sqrt{3}} - 1\right)R = 0.155R$

(ii) $AB = \sqrt{2}a = 2R$

$AC = \frac{\sqrt{3}\,a}{2} = R + r$

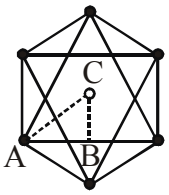



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Now, $\frac{R+r}{R}=\frac{\sqrt{3}}{\sqrt{2}}$

$\therefore \quad r=\left(\sqrt{\frac{3}{2}}-1\right)R=0.225R$

(iii) $AB=R$



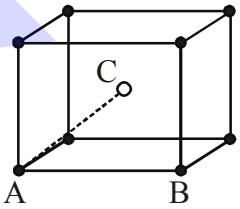

$AC=R+r$

$\angle BAC=45°$

Now, $\cos 45°=\frac{AB}{AC}\Rightarrow\frac{1}{\sqrt{2}}=\frac{R}{R+r}$

$\therefore \quad r=\left(\sqrt{2}-1\right)R=0.414R$

(iv) $AB=a=2R$




$AC=\frac{\sqrt{3}a}{2}=R+r$

Now, $\frac{R+r}{R}=\sqrt{3}$

$\therefore \quad r=\left(\sqrt{3}-1\right)R=0.732R$

***Ex.8*** ***An element (molar mass = 60 gm/mole) crystallises in CCP lattice. If density is 6.25 gm/cm$^3$ and the distance between next nearest neighbour is dÅ, then the value of 'd' is ($N_A = 6 \times 10^{23}$).***

***Ex.8*** **Ans.(4)**

$$d=\frac{ZM}{N_A.a^3}$$

$$6.25=\frac{4\times 60}{6\times 10^{23}\times a^3}\Rightarrow a=4\times 10^{-8}\text{cm}=4\text{Å}.$$

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## 10 PACKING IN IONIC SOLID

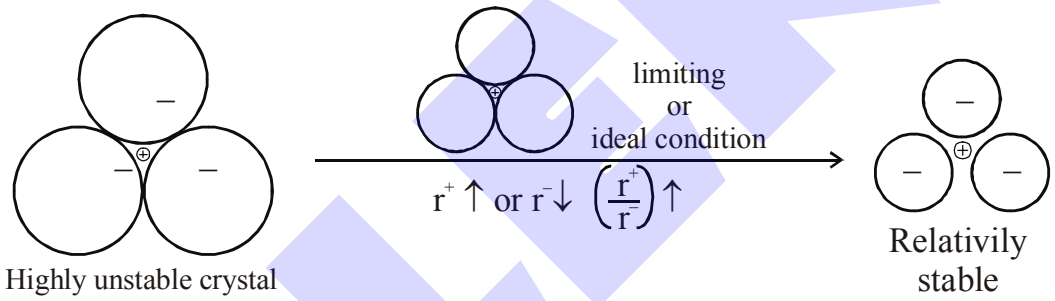
**(i)** Normally anions are bigger than cations, hence ionic solid are considered as the packing of anions and cations are supposed to occupy the voids.

**(ii)** Oppositively charged particles should lie closer and smilarly charged particles should lie away from each other.

**(iii)** Each ion tend to maximise its coordination number (Number of oppositively charged ions around it).

Point (ii) and (iii) contradict each other because on increase the CN, the repulsion between like charges will also increases. Hence, in all the ionic solids, there must be a balance with the help of relative size of ions to ensure maximum CN and minimum repulsion between like charges.

### 10.1 LIMITING OR IDEAL RADIUS RATION $\left(\frac{r^+}{r^-}\right)$:




The minimum $\frac{r^+}{r^-}$ values for the existence of a cation in a particular void is called **limiting radius ratio** for that void.

With the increase in radius ratio, space between anion will increase & hence the cation may tend for higher coordination number.

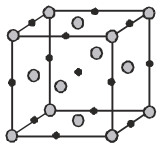
| Voids | CN | Limiting $r^+/r^-$ | Range of $r^+/r^-$ |
|---|---|---|---|
| Triangular | 3 | 0.155 | 0.155 – 0.225 |
| Tetrahedral | 4 | 0.225 | 0.225 – 0.414 |
| Octahedral | 6 | 0.414 | 0.414 – 0.732 |
| Cubic | 8 | 0.732 | 0.732 – 1.000 |

* For values lesser than 0.155, crystal will not exist.

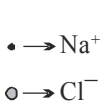
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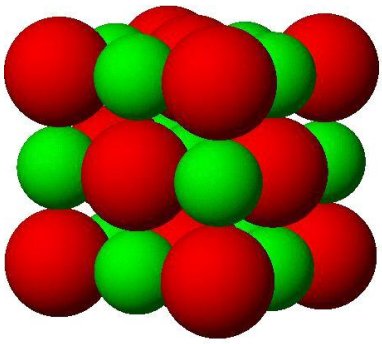
## 10.2 *TYPES OF IONIC STRUCTURES:*

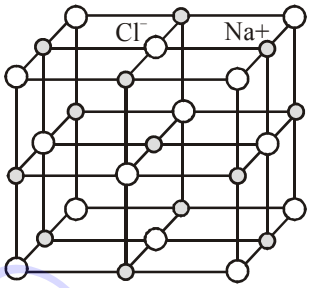
### 10.2.1 Rock salt structure (NaCl) :



Rock salt structure






The bigger $Cl^-$ forms cubic close packing and small $Na^+$ occupy positions of all octahedral voids. The radius ratio $\frac{r^+}{r^-}$ lie in the range 0.414 – 0.732.

**(i)** Each $Na^+$ is surrounded by six $Cl^-$ and each $Cl^-$ is surrounded by six $Na^+$ ion. [6:6 coordination]

**(ii)** No. of $Na^+$ and $Cl^-$ in each unit cell is 4.

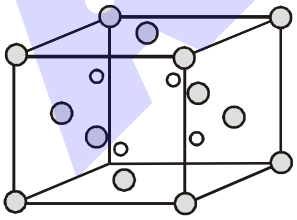
**(iii)** Number of formula units of NaCl per unit cell is equal to 4.

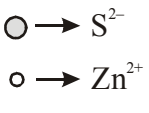
**(iv)** The density of NaCl crystal is given by $d = \left(\frac{4 \times M_{NaCl}}{N_A \times a^3}\right)$

**(v)** The edge length of NaCl unit cell is given by $(2r^+ + 2r^-) \Rightarrow \boxed{\frac{a}{2} = r^+ + r^-}$

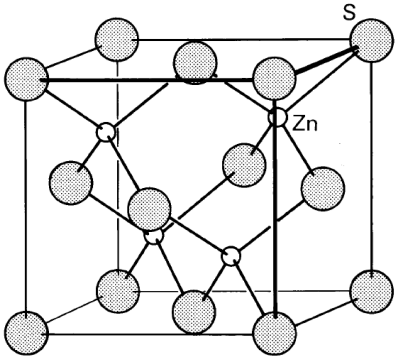
### 10.2.2 Zinc blende (sphalerite) structure (ZnS) :

Larger ion ($S^{2-}$) forming ccp arrangement and smaller ion ($Zn^{2+}$) filling half or alternate tetrahedral voids






Zinc blende structure




**(i)** C.N. of $Zn^{2+} = 4$ ; C.N. of $S^{2-} = 4$ [4 : 4 coordination]

**(ii)** Formula units of ZnS per unit cell = 4. **(iii)** $d_{ZnS} = \frac{4 \times M_{ZnS}}{N_A \times a^3}$ **(iv)** $r_{Zn^{2+}} + r_{S^{2-}} = \frac{a\sqrt{3}}{4}$

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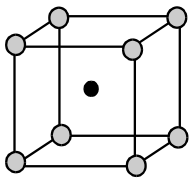
### 10.2.3 Cesium chloride structure (CsCl) :

$Cl^-$ at the corners of cube and $Cs^+$ in the center (cubic void).

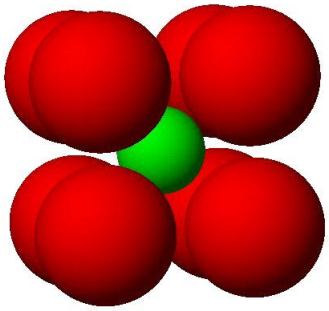
**(i)** C.N. of $Cs^+ = 8$ ; C.N. of $Cl^- = 8$ [8 : 8 coordination]

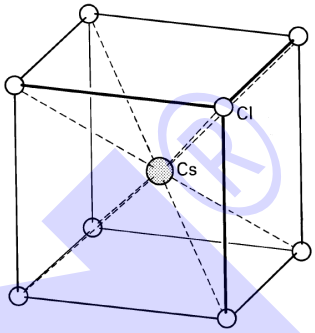
**(ii)** Formula units of CsCl per unit cell = 1

**(iii)** $d_{CsCl} = \frac{M_{CsCl}}{N_A \times a^3}$ **(iv)** $r_{Cs^+} + r_{Cl^-} = \frac{a\sqrt{3}}{2} \Rightarrow \boxed{r^+ + r^- = \frac{a\sqrt{3}}{2}}$



Cesium chloride structure





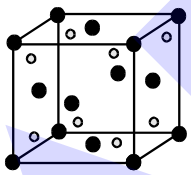

### 10.2.4 Fluorite structure ($CaF_2$) :

$Ca^{2+}$ forming ccp arrangement and $F^-$ filling all tetrahedral voids.

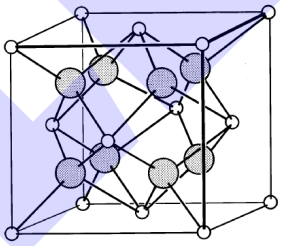
**(i)** C.N. of $F^- = 4$ ; C.N. of $Cs^+ = 8$ [8 : 4 coordination]

**(ii)** Formula units of $CaF_2$ per unit cell = 4

**(iii)** $d_{CaF_2} = \frac{4 \times M_{CaF_2}}{N_A \times a^3}$ **(iv)** $r_{Ca^{2+}} + r_{F^-} = \frac{a\sqrt{3}}{4}$



Fluorite structure



### 10.2.5 Antifluorite structure ($Li_2O$) :

$O^{2-}$ ion forming ccp and $Li^+$ taking all tetrahedral voids.

**(i)** C.N. of $Li^+ = 4$

C.N. of $O^{2-} = 8$

**(ii)** Formula units of $Li_2O$ ; per unit cell = 4

**(iii)** $d_{Li_2O} = \frac{4 \times M_{Li_2O}}{N_A \times a^3}$ **(iv)** $r_{Li^+} + r_{O^{2-}} = \frac{a\sqrt{3}}{4}$

### 10.2.6 Corundum Structure ($Al_2O_3$) :

$O^{2-}$ forming hcp and $Al^{3+}$ filling 2/3 octahedral voids.

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**10.2.7 Rutile structure** ($TiO_2$) **:**

$O^{2-}$ forming hcp while $Ti^{4+}$ ions occupy half of the octahedral voids.

**10.2.8 Pervoskite structure** ($CaTiO_3$) **:**

$Ca^{2+}$ in the corner of cube, $O^{2-}$ at the face center and $Ti^{4+}$ at the centre of cube.

**10.2.9 Spinel and inverse spinel structure** ($MgAl_2O_4$) **:**

$O^{2-}$ forming fcc, $Mg^{2+}$ filling 1/8 of tetrahedral voids and $Al^{3+}$ taking half of octahedral voids.

In an inverse spinel structure, $O^{2-}$ ion form FCC lattice, $A^{2+}$ ions occupy 1/4 of the octahedral voids and trivalent cation occupies 1/8 of the tetrahedral voids and 1/4 of the octahedral voids.

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***Ex.9 A solid $A^+B^-$ has NaCl type close packed structure. If the anion has a radius of 250 pm, what should be the ideal radius for the cation ? Can a cation $C^+$ having a radius of 180 pm be slipped into the tetrahedral site of the crystal $A^+B^-$ ? Give reason for your answer.***

***Sol.*** *In $Na^+Cl^-$ crystal each $Na^+$ ion is surrounded by 6 $Cl^+$ ions and vice versa. Thus $Na^+$ ion is placed in octahedral hole.*

*The limiting radius ratio for octahedral site = 0.414*

*or* $\frac{A^+}{B^-} = \frac{r}{R} = 0.414$

*Given that radius of anion ($B^-$) R = 250 pm*

*i.e. radius of cation ($A^+$)* $r = 0.414\ R = 0.414 \times 250$ *pm*

*or* $r = 103.5$ *pm*

*Thus ideal radius for cation ($A^+$) is r = 103.5 pm.*

*We know that (r/R) for tetrahedral hole is 0.225.*

$\therefore \quad \frac{r}{R} = 0.225$

***or*** $r = 0.225 \quad R = 0.225 \times 250 = 56.25$ *pm*

*Thus ideal radius for cation is 56.25 pm for tetrahedral hole. But the radius of $C^+$ is 180 pm. It is much larger than ideal radius i.e. 56.25 pm. Therefore we can not slip cation $C^+$ into the tetrahedral site.*

***Ex.10 A compound forms hexagonal close-packed structure. What is the total number of voids in 0.5 mole of it? How many of these are tetrahedral voids?***

***Sol.*** *Since, for every atom forming hcp structure there are two tetrahedral voids and one octahedral void. Total voids = 3 × 0.5 = 1.5 mol and total tetrahedral void = 2 × 0.5 = 1 mol.*

***Ex.11 A compound is formed by two elements M and N. The element N forms ccp and atoms of M occupy 1/3rd of tetrahedral voids. What is the formula of the compound?***

***Sol.*** *In one unit cell of ccp, no. of N = 4 ; total no. of tetrahedral void = 8 ; occupied tetrahedral void by M = 8/3 ; Empirical formula : $N_4M_{8/3} = N_3M_2$.*

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## 11. IMPERFECTIONS IN SOLIDS

Although crystalline solids have short range as well as long range order in the arrangement of their constitutent particles, yet crystals are not perfect. Usually a solid consists of an aggregate of large number of small crystals. These small crystals have defects in them. This happens when crystallisation process occurs at fast or moderate rate. Single crystals are formed when the process of crystallisation occurs at extremely slow rate. Even these crystals are not free of defects.

The defects are basically irregularities in the arrangement of constituent particles. Broadly speaking, the defects are of two types, namely, **point defects** and **line defects**. Point defects are the irrgularties or deviations from ideal arrangement around a point or an atom in a crystalline substance, whereas the line defects are the irregularities or deviations from ideal arrangement in entire rows of lattice points. These irregularities are called crystal defects. We shall confine our discussion to point defects only.

### 11.1 Types of Point Defects

Point defect can be classified into three types:

(i) Stoichiometric defects

(ii) Non-stoichiometric defects

(iii) Impurity added defect

### (i) Stoichiometric Defect

These are the point defect that do not disturb the stoichiometry of the solid. They are also called ***intrinsic*** or **thermodynamic defects**. Basically these are of two types; vacancy defects and interstitial defect.

**(a) Vacancy Defect :**

When some of the lattice sites are vacant, the crystal is said to have vacancy defect. This results in decrease in density of the substance. This defect can also develop when a substance is heated.

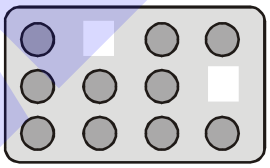


**Fig. :** Vacancy defects

**(b) Interstitial Defect :** When some constituent particles (atoms or molecules) occupy an interstitial site. the crystal is said to have interstitial defect. This defect increases the density of the substance.

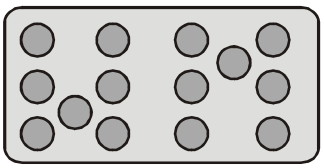


**Fig. :** Interstitial defects

Vacancy and interstitial defects as explained above can be shown by non-ionic solids. Ionic solids must always maintain electrical neutrality. Rather than simple vacancy or interstitial defects, they show these defects as **Frenkel and Schottky defects**.

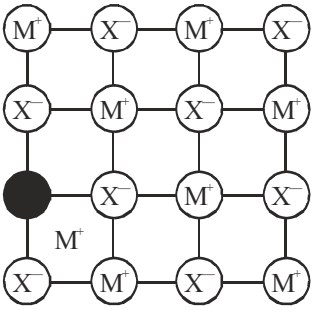
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**(c) Frenkel Defect :** This defect is shown by ionic solids.

The smaller ion (usually cation) is delocalised from its normal site to an interstitial site . It creates a vacancy defect at its original site and an interstitial defect at its new location.

Frenkel defect is also called **dislocation defect**. It does not change the density of the solid.

Frenkel defect is shown by ionic substance in which there is a large difference in the size of ions, for example, ZnS, AgCl, AgBr and AgI due to small size of $Zn^{2+}$ and $Ag^{+}$ ions.



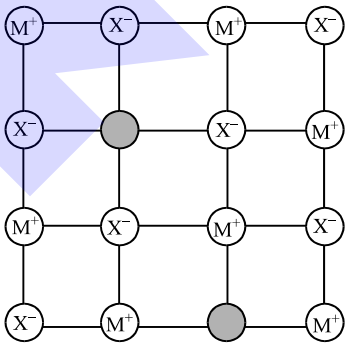
Frenkel Defect

**Influences :** Makes solid crystals good conductor. In Frenkel defect , ions in interstitial sites increases the dielectric constant.

**(d) Schottky Defect :** It is basically a vacancy defect in ionic solids. In order to maintain electrical neutrality, the number of missing cations and anions are equal (in stoichiometric ratio)

Like simple vacancy defect, Schottky defect also decreases the density of the substance. Number of such defects in ionic solids is quite significant. For example, in NaCl there are approximately,

$10^6$ Schottky pairs per $cm^3$ at room temperature. In 1 $cm^3$ there are about $10^{22}$ ions. Thus, there is one Schottky defect per $10^{16}$ ions. Schottky defect is shown by ionic substances in which the cation and anion are of almost similar sizes. For example, NaCl, KCl, CsCl and AgBr. It may be noted that AgBr shows both, Frenkel as well as Schottky defects.



Schottky Defect

**Influence :** The presence of large number of schottky defects in crystal results in significant decrease in its density.

**(ii) Non-Stoichiometric Defects**

The defects discussed so far do not disturbs the stoichiometry of the crystalline substance. However, a large number of non-stoichiometric inorganic solids are known which contain the constituent elements in non-stoichiometric ratio due to defects in their crystal structures. These defects are of two types:

(a) Metal excess defect.

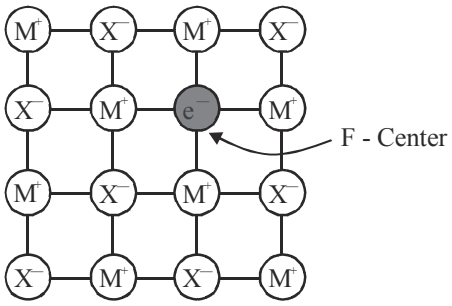
(b) Metal deficiency defect.

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**(a) Metal Excess Defect**

**(I) Metal excess defect due to anionic vacancies :**

Alkali halides like NaCl and KCl show his type of defect. When crystals of NaCl are heated in an atmosphere of sodium vapour, the sodium atoms are deposited on the surface of the crystal. The $Cl^-$ ions diffuse to the surface of the crystal and combine with Na atoms to give NaCl. This happens by loss of electron by sodium atoms to form $Na^+$ ions.



Metal excess defects due to anion vacancies

The released electrons diffuse into the crystal and occupy anionic sites . As a result the crystal now has an excess of sodium. The anionic sites occupied by unpaired electrons are called ***F-centres*** (from the German word ***Farbenzenter*** for colour centre).

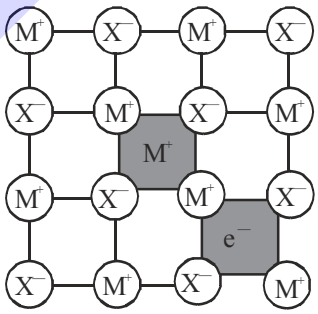
They impart yellow colour to the crystals of NaCl. The colour results by excitation of these electrons when they absorb energy from the visible light falling on the crystals. Similarly, excess of lithium makes LiCl crystals pink and excess of potassium makes KCl crystals violet (or lilac).

**(II) *Metal excess defect due to the presence of extra cations at interstitial sites :***

Zinc oxide is white in colour at room temperature. On heating it loses oxygen and turns yellow.

$$ZnO \xrightarrow{\text{heating}} Zn^{2+} + \frac{1}{2}O_2 + 2e^-$$

Now there is excess of zinc in the crystal and its formula becomes $Zn_{1+x}O$. The excess $Zn^{2+}$ ions move to interstitial sites and the electrons to neighbouring interstitual sites.


Metal excess defects due to interstitial cation

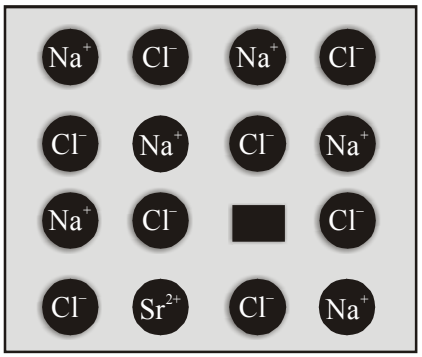
**(b) Metal Deficiency Defect**

There are many solids which are difficult to prepare in the stoichiometric composition and contain less amount of the metal as compared to the stoichiometric proportion. A typical example of this type is FeO which is mostly found with a composition of $Fe_{0.95}O$. It may actually range from $FeO_{0.93}O$ to $Fe_{0.96}O$. In crystals of FeO, some $Fe^{2+}$ cations are missing and the loss of positive charge is made up by the presence of required number of $Fe^{3+}$ ions.

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**(iii) Impurity Defects**

If molten NaCl containing a little amount of $SrCl_2$ is crystallised, some of the sites of $Na^+$ ions are occupied by $Sr^{2+}$. Each $Sr^{2+}$ replaces two $Na^+$ Ions. It occupies the site of one ion and the other site remains vacant. The cationic vacancies thus produced are equal in number to that of $Sr^{2+}$ ions. Another similar example is the solid solution of $CdCl_2$ and AgCl.



Introduction of cation vacancy in NaCl by substitution of $Na^+$ by $Sr^{2+}$

**Note:** (i) As temperature increases, no. of defects increases exponentially.

(ii) For defect formation : $\Delta H > 0$ , $\Delta S > 0$

$\therefore$ More spontaneous at higher temperatures.

(iii) No matter how many imperfections are present in a crystal, it is always electrically neutral (no net charge).

## 12. ELECTRICAL PROPERTIES

Solids exhibit an amazing range of electrical conductivities, extending over 27 orders of magnitude ranging from $10^{-20}$ to $10^7$ $ohm^{-1}$ $m^{-1}$. Solids can be classified into three types on the basis of their conductivities.

**(i) Conductors :** The solids with conductivities ranging between $10^4$ to $10^7$ $ohm^{-1}m^{-1}$ are called conductors. Metals having conductivities in the order of $10^7$ $ohm^{-1}m^{-1}$ are good conductors.

**(ii) Insulators :** These are the solids with very low conductivities ranging between $10^{-20}$ to $10^{-10}$ $ohm^{-1}m^{-1}$.

**(iii) Semiconductors :** These are the solids with conductivities in the intermediate range from $10^{-6}$ to $10^4$ $Ohm^{-1}m^{-1}$.

### 12.1 : Conduction of Electricity in Metals

A conductor may conduct electricity through movement of electrons or ions. Metallic conductors belong to the former category and electrolytes to the latter.

Metals conduct electricity in solid as well as molten state. The conductivity of metals depend upon the number of valence electrons available per atom. The atomic orbitals of metal atoms from molecular orbitals which are so close in energy to each other as to form a band. If this band is partially filled or it overlaps with a higher energy unoccupied conduction band, then electrons can flow easily under an applied electric field and the metal shows conductivity.

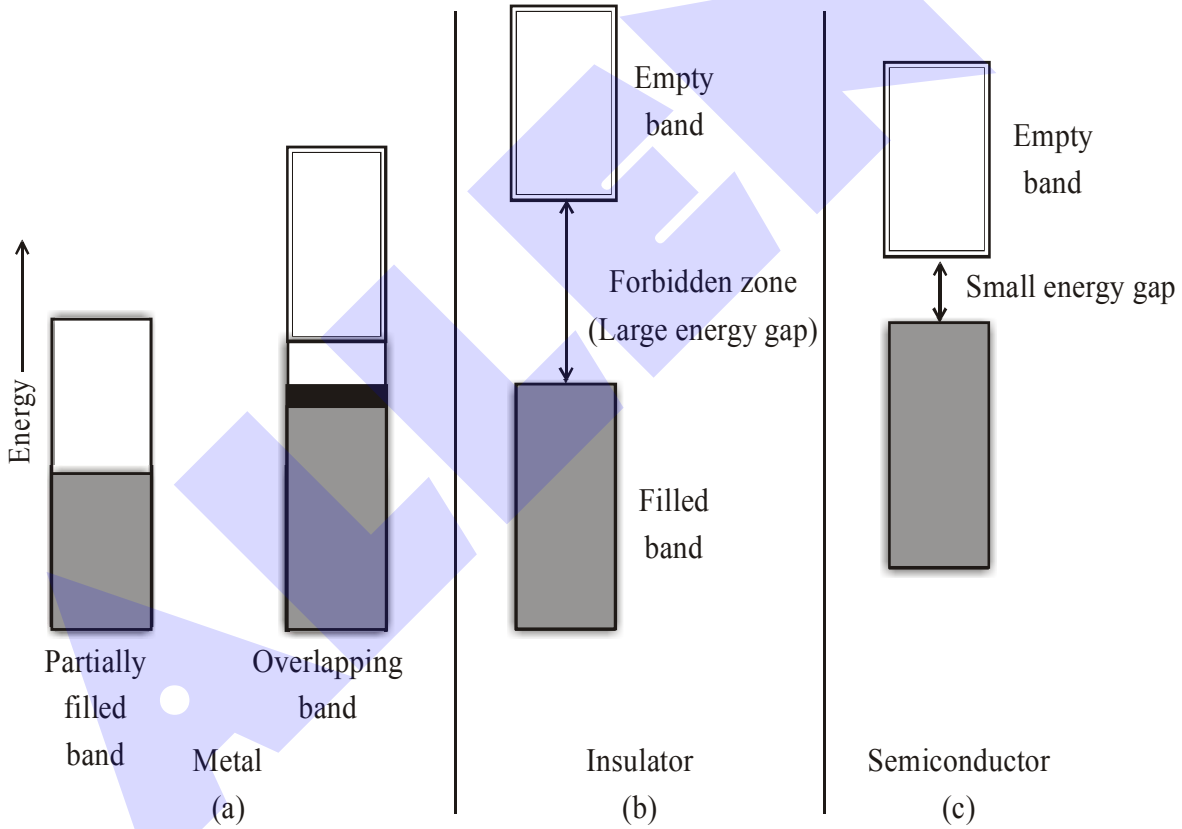
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### 12.2 : Insulator :

If the gap between filled valence band and the next higher unoccupied band (conduction band) is large, electron cannot jump to it and such a substance has very small conductivity and it behaves as an insulator.

### 12.3 : Conduction of Electricity in Semi-conductor

In case of semiconductors, the gap between the valence band and conduction band is small. Therefore, some electrons may jump to conduction band and show some conductivity. Electrical conductivity of semconductors increases with rise in temperature, since more electrons can jump to the conduction band. Substances like silicon and germanium show this type of behaviour and are called **intrinsic semiconductors**.




**Distinction among (a) metals (b) insulators and (c) semiconductors. In each case, an unshaded area represents a conduction band.**

The conducitivity of these interinsic semiconductors is too low to be of practical use. Their conductivity is increased by adding an appropriate amount of suitable impurity. This process is called **doping**. Doping can be done with an impurity which is electron rich or electron deficient as compared to the intrinsic semiconductor silicon or germanium. Such impurities introduce ***electrical defect*** in them.

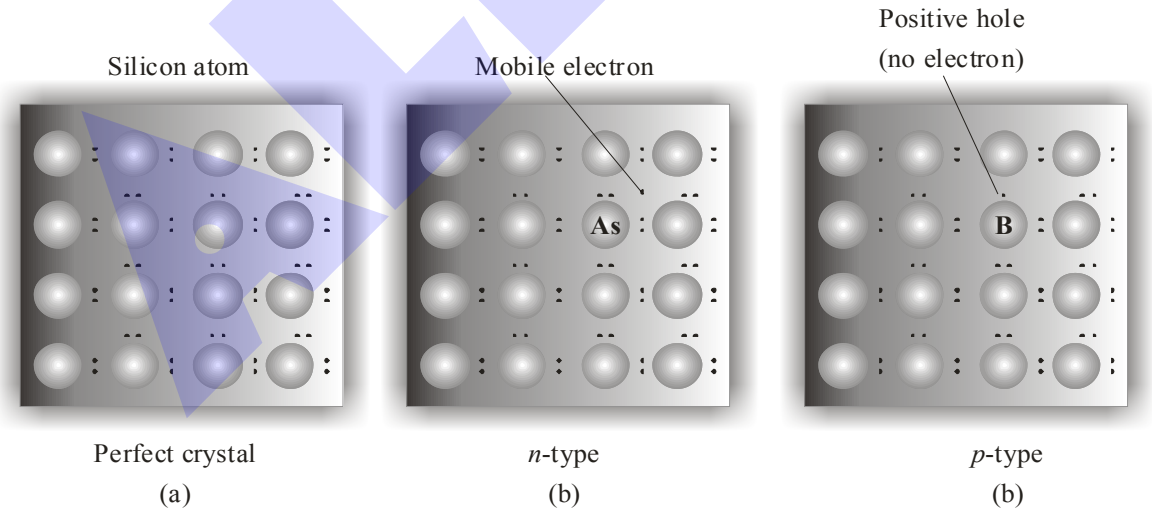
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**(a) Electron – rich impurities**

Silicon and germanium belong to group 14 of the periodic table and have four valence electrons each. In their crystals each atom forms four covalent bonds with its neighbours . When doped with a group 15 element like P or As, which contains five valence electrons, they occupy some of the lattice sites in silicon or germanium crystal. Four out of five electrons are used in the formation of four covalent bonds with the four neighbouring silicon atoms. The fifth electron is extra and becomes delocalised. These delocalised electrons increase the conductivity of doped silicon (or germanium). Here the increase in conductivity is due to the negatively charged electron, hence silicon doped with electron-rich impurity is called ***n-type semiconductor***.

**(b) Electron – deficit impurities**

Silicon or germanium can also be doped with a group 13 element like B, Al or Ga which contains only three valence electrons. The place where the fourth valence electron is missing is called ***electron hole or electron vacancy*** . An electron from a neighbouring atom can come and fill the electron hole, but in doing so it would leave an electron hole at its original position. If it happens, it would appear as if the electron hole has moved in the direction opposite to that of the electron that filled it. Under the influence of electric field, electrons would move towards the positively charged plate through electronic holes, but it would appear as if electron holes are positively charged and are moving towards negatively charged plate. This type of semi conductors are called ***p-type semiconductors***.




***Creation of n-type and p-type semi conductors by doping groups 13 and 15 elements***

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**(c) Applications of n-type and p-type semiconductors**

Various combinations of *n*-type and *p*-type semiconductors are used for making electronic components. Diode is a combination of *n*-type and *p*-type semiconductor and is used as a rectifier. Transistors are made by sandwiching a layer of one type of semiconductor between two layers of the other type of semiconductor. *npn* and *pnp* type of transistors are used to detect or amplify radio or audio signals. The solar cell is an efficient photo-diode used for conversion of light energy into electrical energy.

Germanium and Silicon are group 14 elements and therefore, have a characteristic valency of four and form four bonds as in diamond. A large variety of solid state materials have been prepared by combination of group 13 and 15 or 12 and 16 to simulate average valence of four as in Ge or Si. Typical compounds of groups 13-15 are InSb, AIP and GaAs. Gallium arsenide (GaAs) semiconductor have very fast response and have revolutionised the design of semiconductor devices. ZnS, CdS, CdSe and HgTe are examples of groups 12-16 compounds. In these compounds, the bonds are not perfectly covalent and the ionic character depends on the electronegativities of the two elements.

It is interesting to learn that transition metal oxides show marked differences in electrical properties. TiO, $CrO_2$ and $ReO_3$ behave like metals. Rhenium oxide, $ReO_3$ is like metallic copper in its conductivity and appearance. Certain other oxides like VO, $VO_2$, $VO_3$ and $TiO_3$ show metallic or insulating properties depending on temperature.

## 13. MAGNETIC PROPERTIES :

Every substance has some magnetic properties associated with it. The origin of these properties lies in the electrons. Each electron in an atom behaves like a tiny magnet. Its magnetic moment originates from two types of motions (i) its orbital motion around the nucleus and (ii) its spin around its own axis . Electron being a charged particle and undergoing these motions can be considered as a small loop of current which possesses a magnetic moment. Thus, each electron has a permanent spin and an orbital magnetic moment associated with it. Magnitude of this magnetic moment is very small and is measured in the unit called Bohr magneton, $\mu_B$. It is equal to $9.27 \times 10^{-24} A\ m^2$.

On the basis of their magnetic properties, substances can be classified into five categories: (i) paramagnetic (ii) diamagnetic (iii) ferromagnetic (iv) antiferromagnetic and (v) ferrimagnetic.

(i) **Paramagnetism :**

Paramagnetic substances are weakly attracted by a magnetic field. They are magnetised in a magnetic field in the same direction. They lose their magnetism in the absence of magnetic field. Paramagnetism is due to presence of one or more unpaired electrons which are attracted by the magnetic field. $O_2$, $Cu^{2+}$, $Fe^{3+}$, $Cr^{3+}$ are some examples of such substances.

(ii) **Diamagnetism :**

Diamagnetic substances are weakly repelled by a magnetic field. $H_2O$, NaCl and $C_6H_6$ are some examples of such substances. They are weakly magnetised in a magnetic field in opposite direction. Diamagnetism is shown by those substances in which all the electrons are paired and there are no unpaired electrons. Pairing of electrons cancels their magnetic moments and they lose their magnetic character.

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(iii) **Ferromagnetism :**

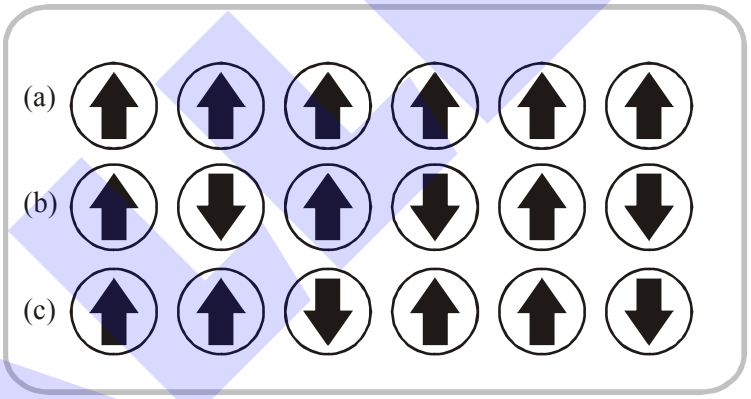
A few substances like iron, cobalt, nickel, gadolinium and $CrO_2$ are attracted very strongly by a magnetic field. Such substances are called ferromagnetic substances. Besides strong attractions, these substances can be permanently magnetised. In solid state, the metal ions of ferromagnetic substances are grouped together into small regions called ***domains***. Thus, each domain acts as a tiny magnet. In an unmagnetised piece of a ferromagnetic substance the domains are randomly oriented and their magnetic moments get cancelled. When the substance is placed in a magnetic field all the domains get oriented in the direction of the magnetic field and a strong magnetic effect is produced. This ordering of domains persist even when the magnetic field is removed and the ferromagnetic substance becomes a permanent magnet.

(iv) **Antiferromagnetism :**

Substances like MnO showing antiferromagnetism have domain structure similar to ferromagnetic substance, but their domains are oppositely oriented and cancel out each other's magnetic moment

(v) **Ferrimagnetism :**

Ferrimagnetism is observed when the magnetic moments of the domains in the substance are aligned in parallel and anti-parallel directions in unequal numbers. They are weakly attracted by magnetic field as compared to ferromagnetic substances. $Fe_3O_4$ (magnetite) and ferrites like $MgFe_2O_4$ and $ZnFe_2O_4$ are examples of such substances. These substances also lose ferrimagnetism on heating and become paramagnetic.




Schematic alignment of magnetic moments in (a) ferromagnetic (b) antiferromagnetic and (c) ferrimagnetic

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## PREVIOUS YEARS SOLVED QUESTION

**1.** A metal cryatallises into two cubic phases, FCC and BCC whose unit cell lengths are 3.5 and 3.0 Å respectively. Calculate the ratio of densities of FCC and BCC. **[JEE-1999]**

**Ans.** **1.259**

**Sol.** $\rho_{Fcc} = \dfrac{4\times M}{N_A\times\left(3.5\times10^{-10}\right)^3}$

$\rho_{Bcc} = \dfrac{2\times M}{N_A\times\left(3\times10^{-10}\right)^3}$

$\dfrac{\rho_{Fcc}}{\rho_{Bcc}} = \dfrac{4}{2}\times\left(\dfrac{3}{3.5}\right)^3$

$= 1.259$

**2.** The coordination number of a metal crystallising in a hcp structure is **[JEE-2000]**

(A) 12 (B) 4 (C) 8 (D) 6

**Ans.** **(A)**

**Sol.** Theory Based

**3.** In any ionic solid [MX] with schottky defects , the number of positive and negative ions are same. **[T/F]** **[JEE-2000]**

**Ans.** **True**

**Sol.** Theory Based

**4.** In a solid "AB" having NaCl structure "A" atoms occupy the corners of the cubic unit cell. If all the face-centred atoms along one of the axes are removed, then the resultant stoichiometry of the solid is

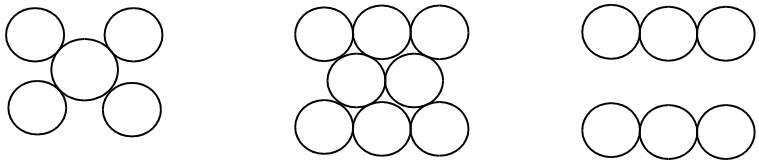
(A) $AB_2$ (B) $A_2B$ (C) $A_4B_3$ (D) $A_3B_4$ **[JEE-2000]**

**Sol.** $\left.\begin{aligned} Z_A &= 8\times\frac{1}{8}+4\times\frac{1}{2}=3\\ Z_B &= 1+12\times\frac{1}{4}=4\end{aligned}\right] \Rightarrow A_3B_4$

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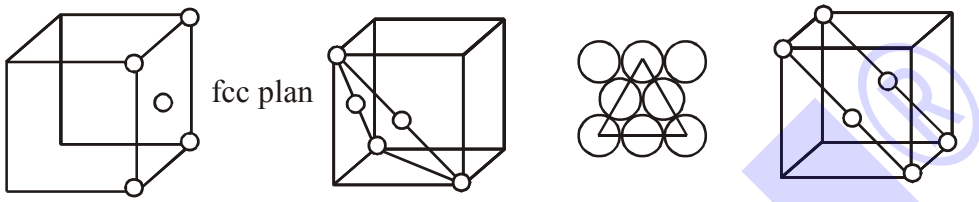
**5.** The figures given below show the location of atoms in three crystallographic planes in FCC lattice. Draw the unit cell for the corresponding structure and identify these planes in your diagram.

**[JEE-2000]**



**Ans.**

**Sol.**




**6.** A substance $A_xB_y$ crystallises in a FCC lattice in which atoms "A" occupy each corner of the cube and atoms "B" occupy the centres of each face of the cube. Identify the correct composition of the substance $A_xB_y$.

(A) $AB_3$ (B) $A_4B_3$

(C) $A_3B$ (D) composition cannot be specified **[JEE-2002]**

**Ans.** **(A)**

**Sol.** $AB_3$

**7.** Marbles of diameter 10 mm each are to be arranged on a flat surface so that their centres lie within the area enclosed by four lines of length each 40 mm. Sketch the arrangement that will give the maximum number of marbles per unit area, that can be enclosed in this manner and deduce the expression to calculate it.

**[JEE 2003]**

**Ans.** 18

**Sol.** $\frac{n \times \pi r^2}{a^2} = \frac{\pi}{2\sqrt{3}}$

$\frac{n \times \pi 5^2}{40 \times 40} = \frac{\pi}{2\sqrt{3}}$

$n = 18.475$

$n \approx 18$

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**8.** (i) AB crystallizes in a rock salt structure with A : B = 1 : 1. The shortest distance between A and B is $Y^{1/3}$ nm. The formula mass of AB is 6.023 Y amu where Y is any arbitrary constant. Find the density in kg $m^{-3}$.

(ii) If measured density is 20 kg $m^{-3}$. Identify the type of point defect. **[JEE–2004]**

**Sol.** (i) Shortest distance b/w A & B = $Y^{1/3}$ nm

$$a = 2Y^{1/3}$$

$$\rho = \frac{4 \times 6.023Y \times 10^{-3}}{6.023 \times 10^{23} \times \left[2Y^{1/3} \times 10^{-9}\right]^3}$$

$$\rho = \frac{4 \times 10^{-3}}{8 \times 10^{-4}}$$

$$= 5 \text{ kgm}^{-3}$$

(ii) copy statement for answer Key

**9.** Which of the following FCC structure contains cations in alternate tetrahedral voids?

(A) NaCl (B) ZnS (C) $Na_2O$ (D) $CaF_2$ **[JEE 2005]**

**Ans.** **(B)**

**10.** An element crystallises in FCC lattice having edge length 400 pm. Calculate the maximum diameter which can be placed in interstital sites without disturbing the structure. **[JEE 2005]**

**Ans.** **117.1 pm**

**Sol.** $\sqrt{2}\, a = 4r$

$$r = \frac{\sqrt{2}a}{4}$$

for octahedral void

$r^1 = .414 \times 141.4$

$d = 2r^1$

$= 117.07$

**11.** The edge length of unit cell of a metal having atomic weight 75 g/mol is 5 Å which crystallizes in cubic lattice. If the density is 2 g/cc then find the radius of metal atom. ($N_A = 6 \times 10^{23}$). Give the answer in pm. **[JEE 2006]**

**Ans.** **216.5 pm**

**Sol.** a = 5Å

M = 75 gram/mol

d = 2 gram/cm

r

$$2 = \frac{Z \times 75}{\left(6.02 \times 10^{+23}\right)\left(5 \times 10^{-8}\right)^3}$$

Z = 2 (Bcc)

$\sqrt{3}\, a = 4r$

$$r = \frac{\sqrt{3}a}{4} = 2.165\text{Å}$$

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**12.** Match the crystal system / unit cells mentioned in Column I with their characteristic features mentioned in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the OMR.

| | **Column I** | **Column II** |
|---|---|---|
| (A) | Simple cubic and face-centred cubic | (P) Have these cell parameters a = b = c and $\alpha = \beta = \gamma$ |
| (B) | Cubic and rhombohedral | (Q) Are two crystal systems |
| (C) | Cubic and tetragonal | (R) Have only two crystallographic angles of 90° |
| (D) | Hexagonal and monoclinic | (S) Belong to same crystal system. |

**[JEE 2007]**

**Ans.** **(A) P, S ; (B) -P,Q ; (C) - Q ; (D) - Q,R**

**Paragraph for Question No. 13 to 15**

In hexagonal systems of crystals, a frequently encountered arrangement of atoms is described as a hexagonal prism. Here, the top and bottom of the cell are regular hexagons and three atoms are sandwiched in between them. A space-filling model of this structure, called hexagonal close-packed (HCP), is constituted of a sphere on a flat surface surrounded in the same plane by six identical spheres as closely as possible. Three spheres are then placed over the first layer so that they touch each other and represent the second layer. Each one of these three spheres touches three spheres of the bottom layer. Finally, the second layer is covered with a third layer that is identical to the bottom layer in relative position. Assume radius of every sphere to be 'r'.

**13.** The number of atoms in this HCP unit cells is **[JEE 2008]**

(A) 4 (B) 6 (C) 12 (D) 17

**Ans.** **(B)**

**Sol.** $12 \times \frac{1}{6} + 2 \times \frac{1}{2} + 3 \times 1 = 6$ atoms

**14.** The volume of this HCP unit cell is **[JEE 2008]**

(A) $24\sqrt{2}\,r^3$ (B) $16\sqrt{2}\,r^3$ (C) $12\sqrt{2}\,r^3$ (D) $\frac{64}{3\sqrt{3}}\,r^3$

**Ans.** **(A)**

**Sol.** $V = \frac{6\sqrt{3}a^3}{4} \times C$

$a = 2r$

$C = \frac{4\sqrt{2}r}{\sqrt{3}}$

$V = 24\sqrt{2}r^3$

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**15.** The empty space in this HCP unit cell is **[JEE 2008]**

(A) 74% (B) 47.6 % (C) 32% (D) 26%

**Ans.** **(D)**

**16.** The correct statement(s) regarding defects in solid is (are) **[JEE 2009]**

(A) Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion.

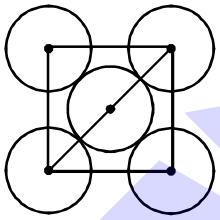
(B) Frenkel defect is a dislocation defect

(C) Trapping of an electron in the lattice leads to the formation of F-center.

(D) Schottky defects have no effect on the physical properties of solids.

**Ans.** **(B,C)**

**17.** The packing effeciency of the two-dimensional square unit cell shown below is **[JEE-2010]**



(A) 39.27% (B) 68.02% (C) 74.05% (D) 78.54%

**Ans.** **(D)**

**Sol.** $PF = \frac{2\pi r^2}{\left(\frac{4r}{\sqrt{2}}\right)^2} = .785$

% PF = 78.5%

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